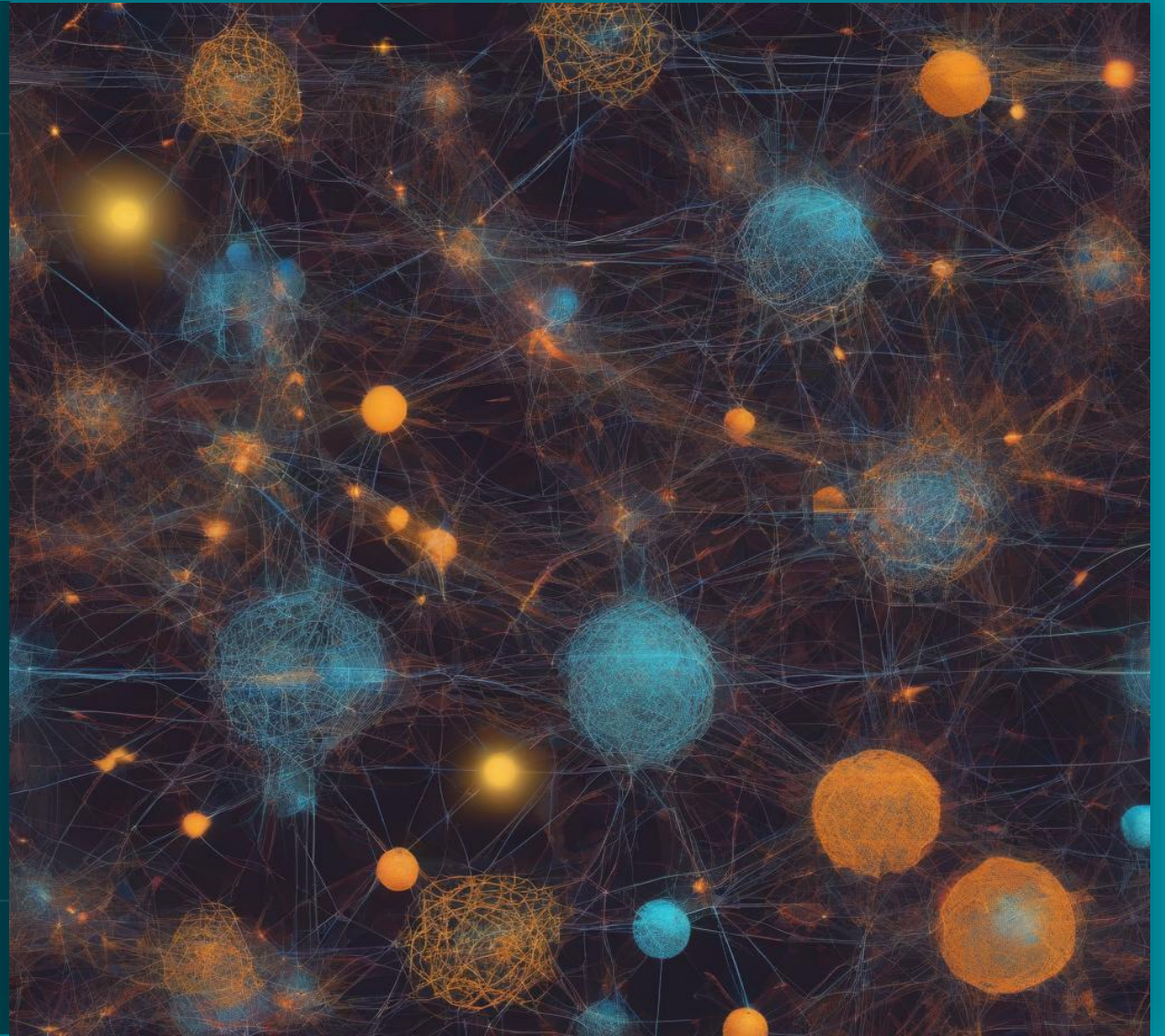


Introduction to Machine Learning in Healthcare

Advanced models

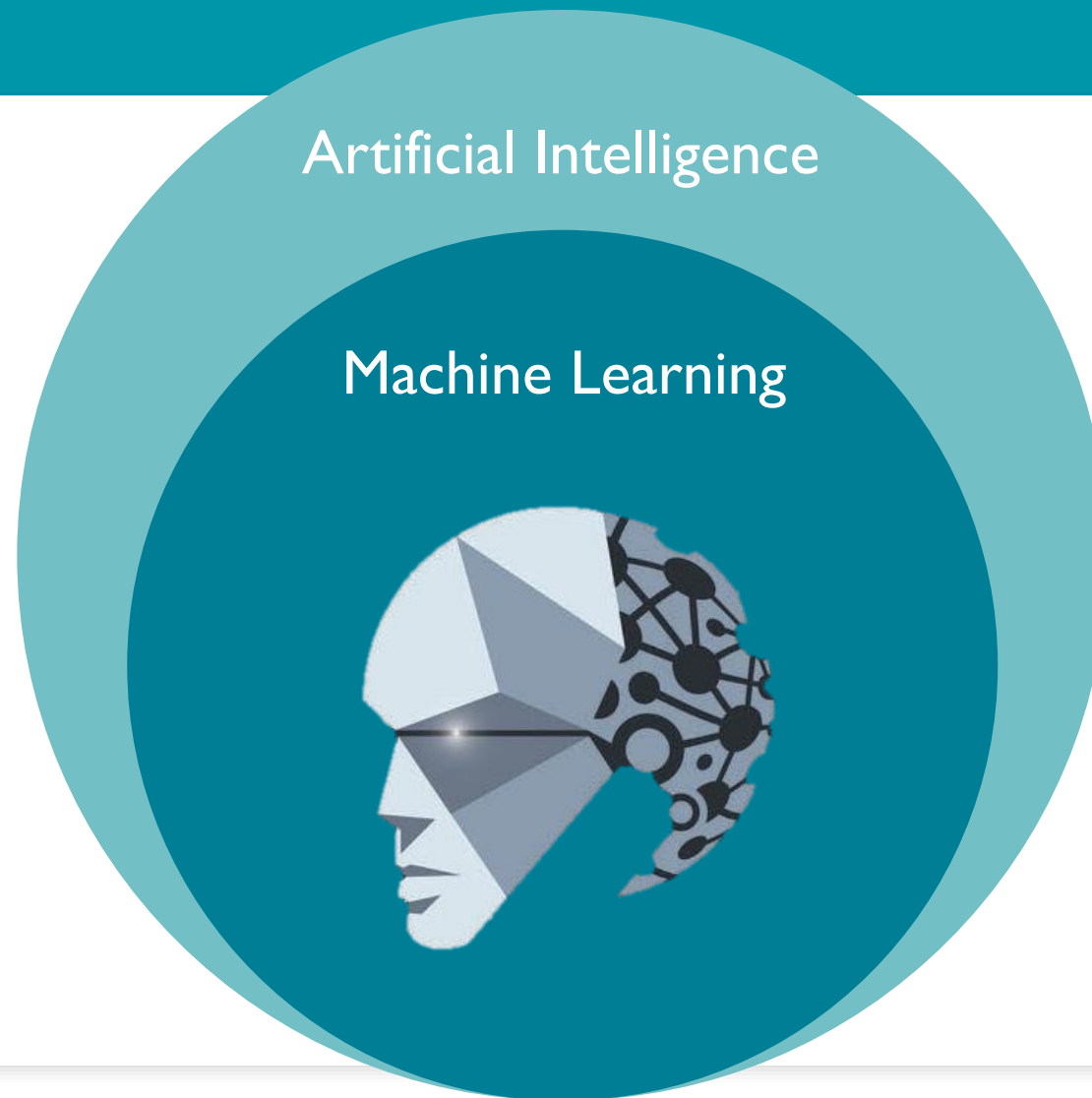


ARTIFICIAL INTELLIGENCE

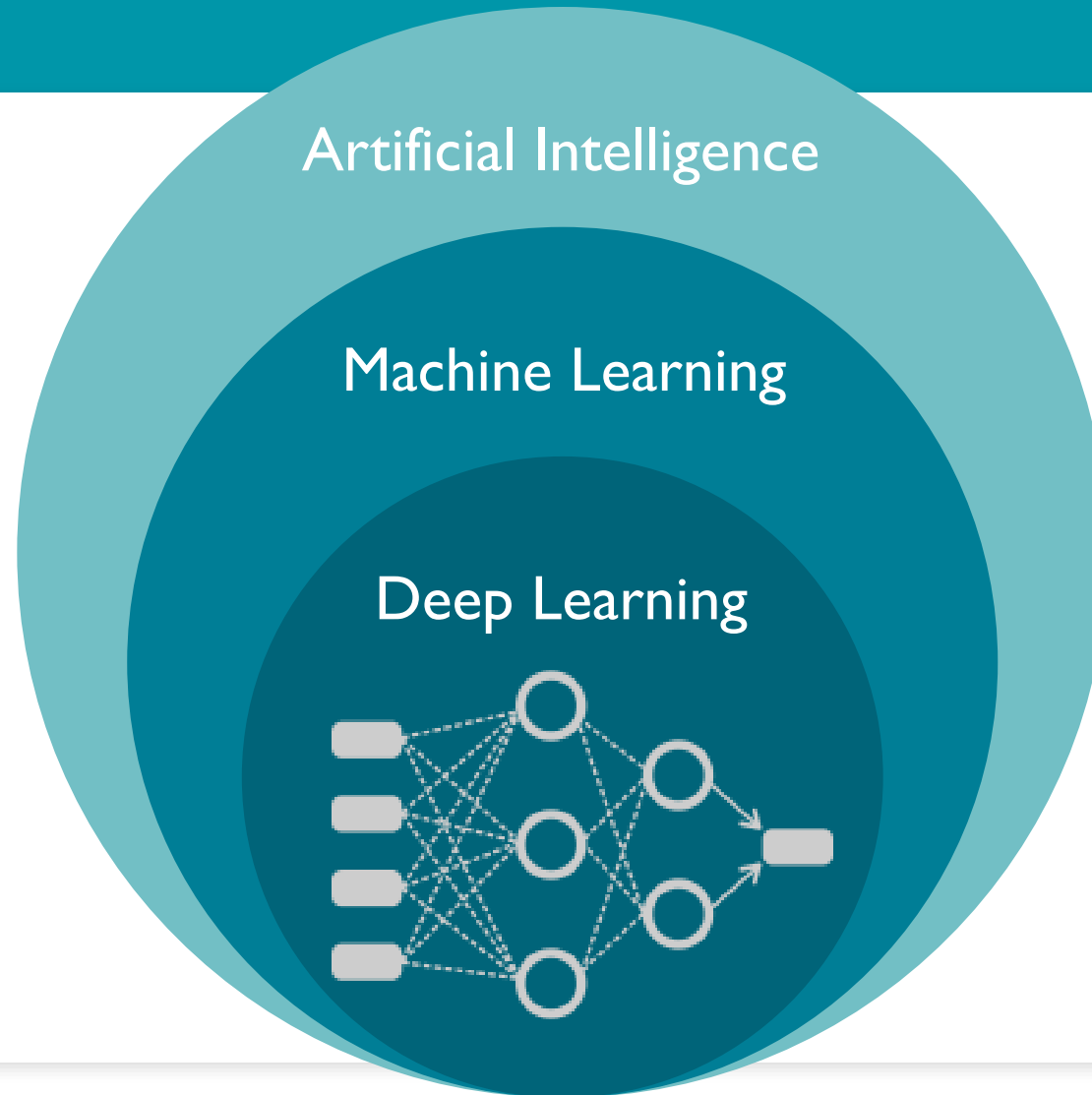
Artificial Intelligence



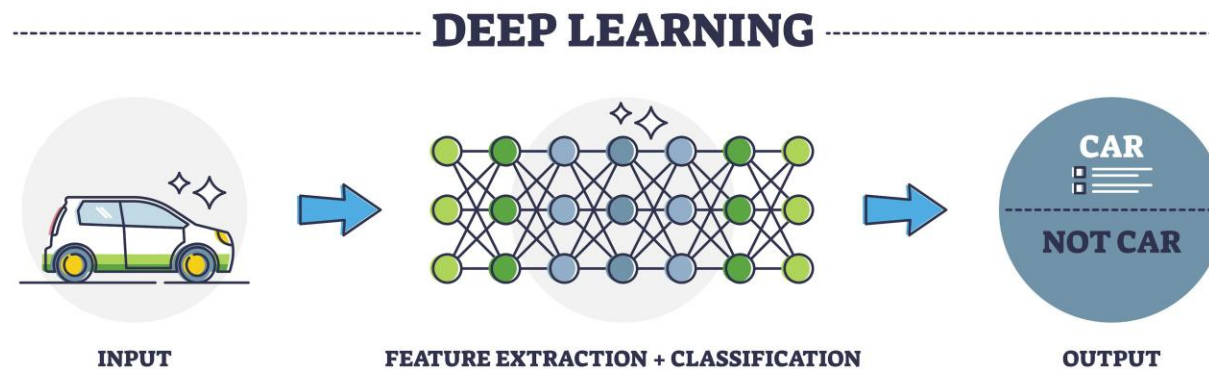
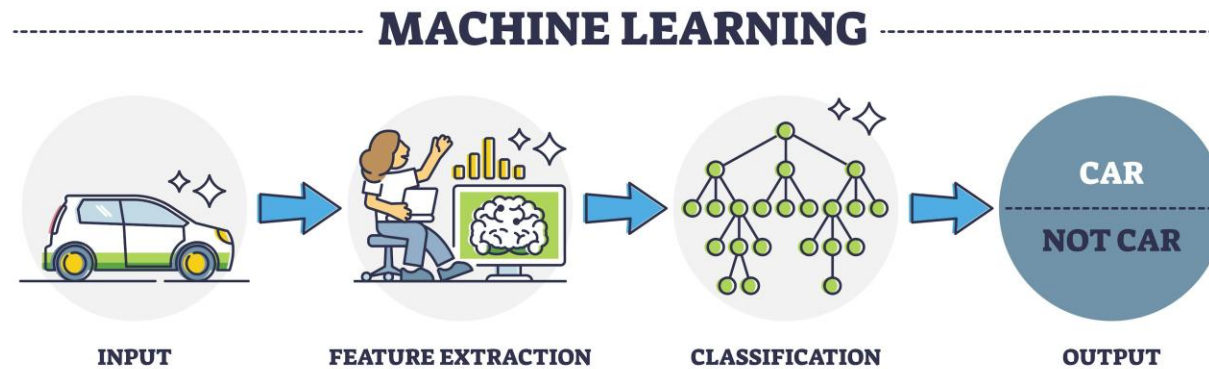
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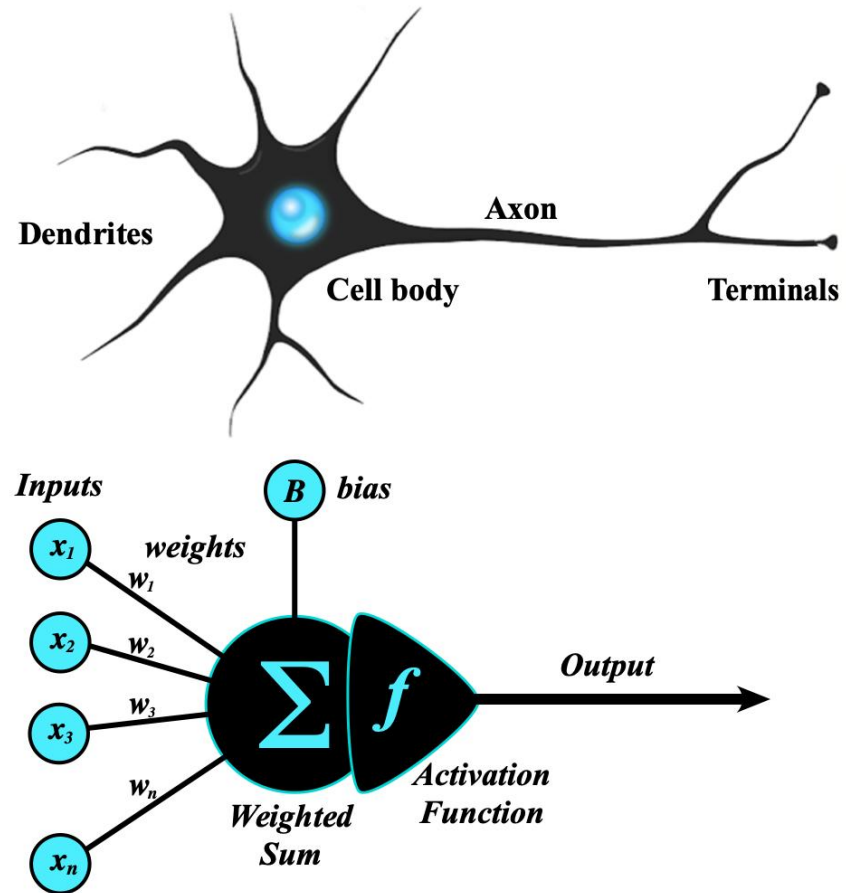
ARTIFICIAL INTELLIGENCE



MACHINE LEARNING VS. DEEP LEARNING

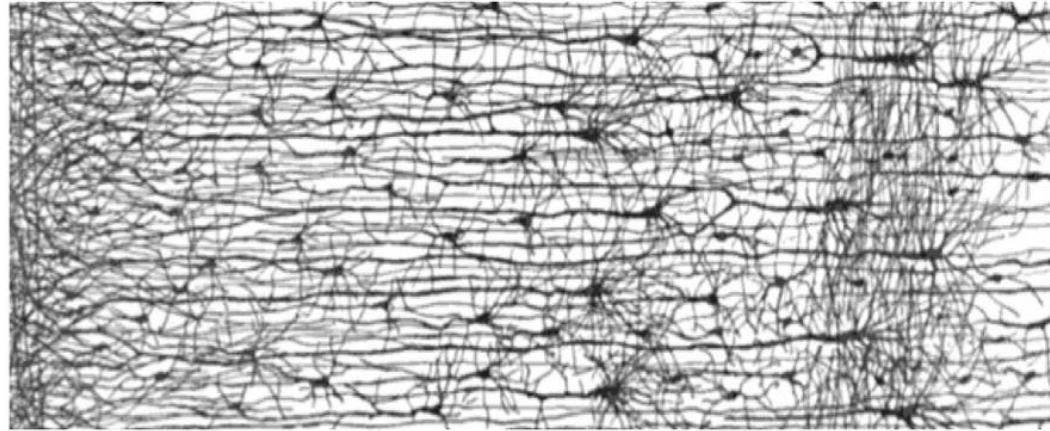


NEURAL NETWORK

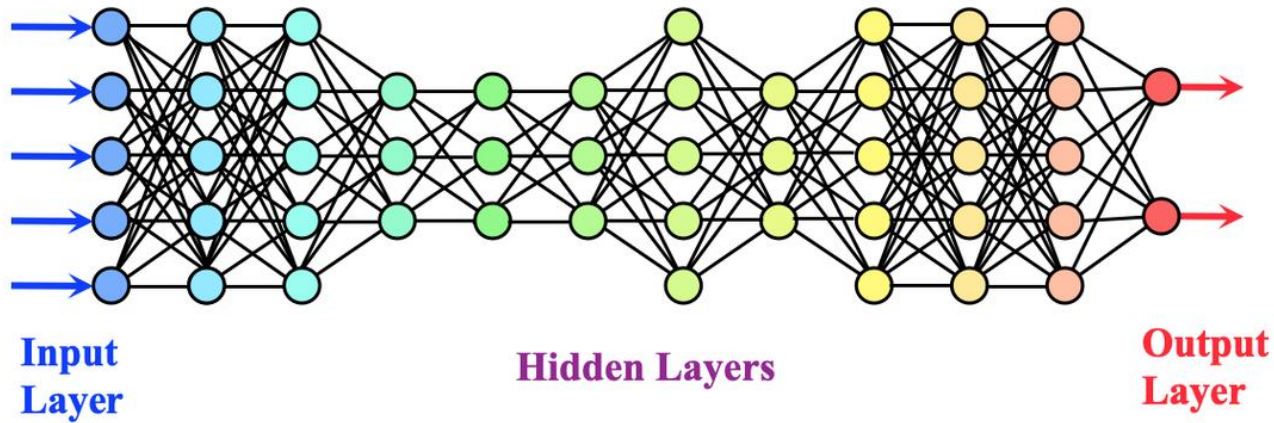


From <https://s.mriquestions.com/what-is-a-neural-network.html>

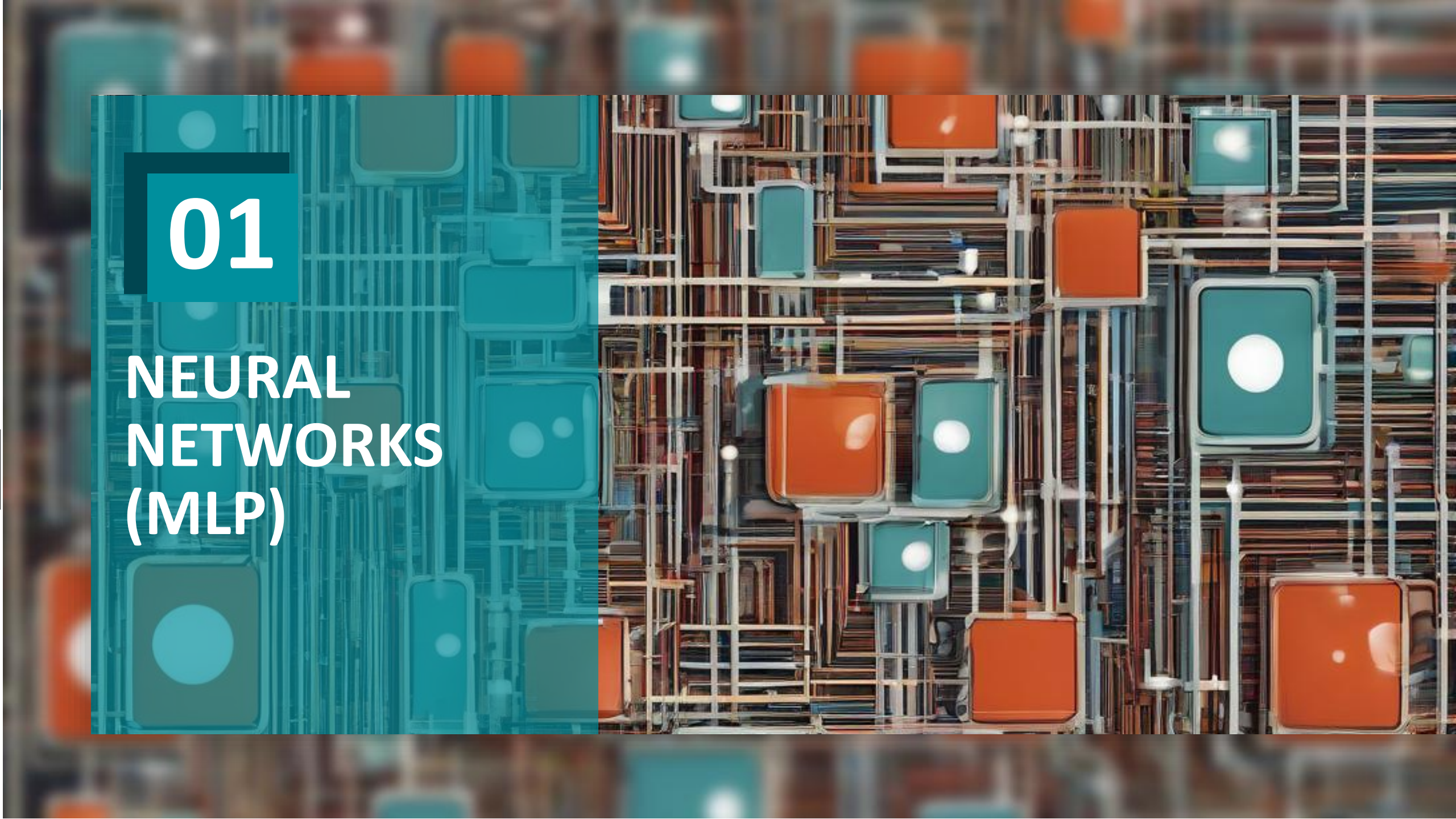
DEEP NEURAL NETWORKS



Santiago Ramón y Cajal



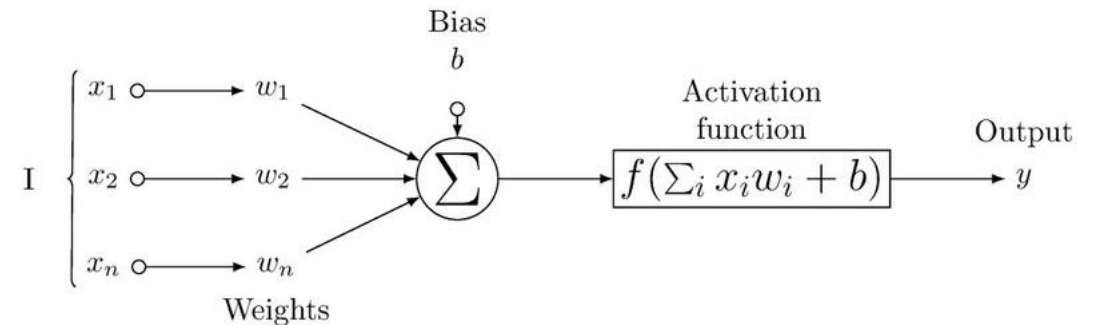
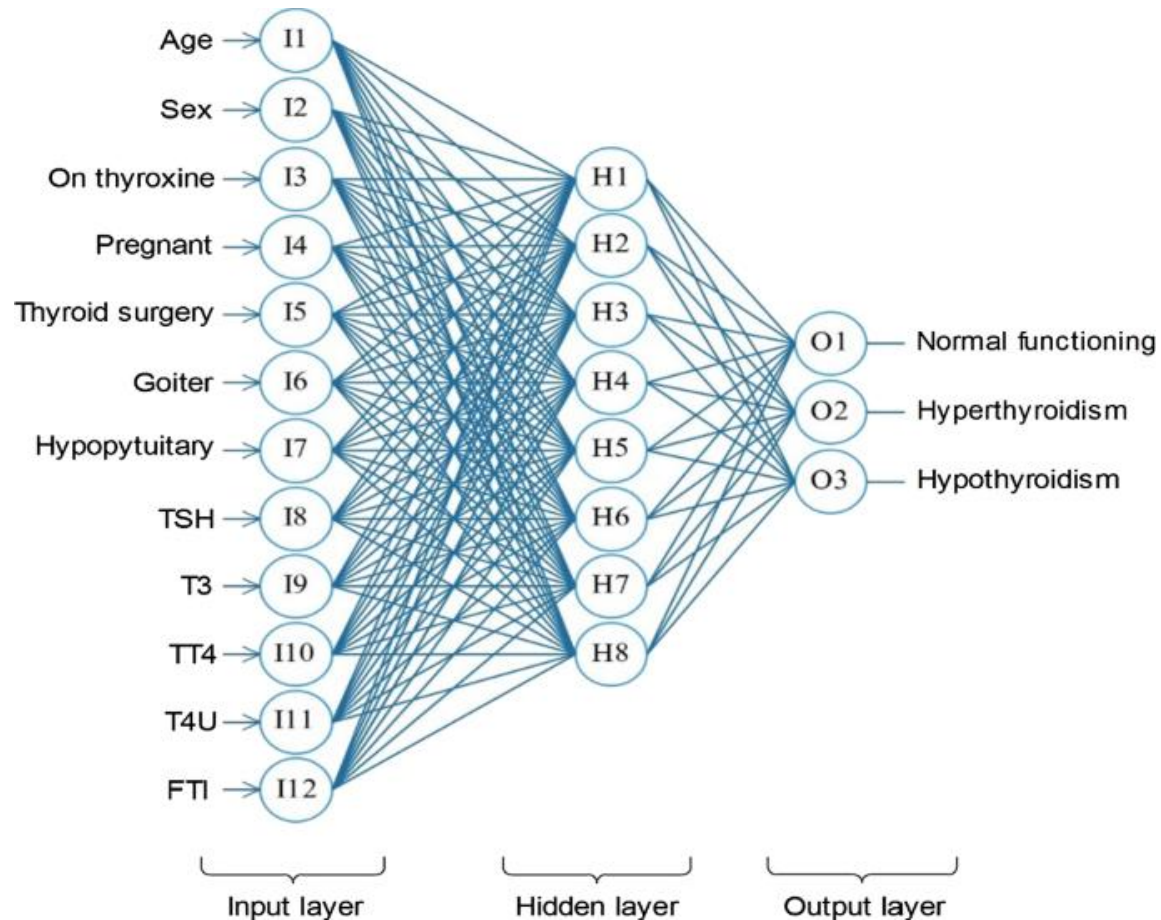
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01

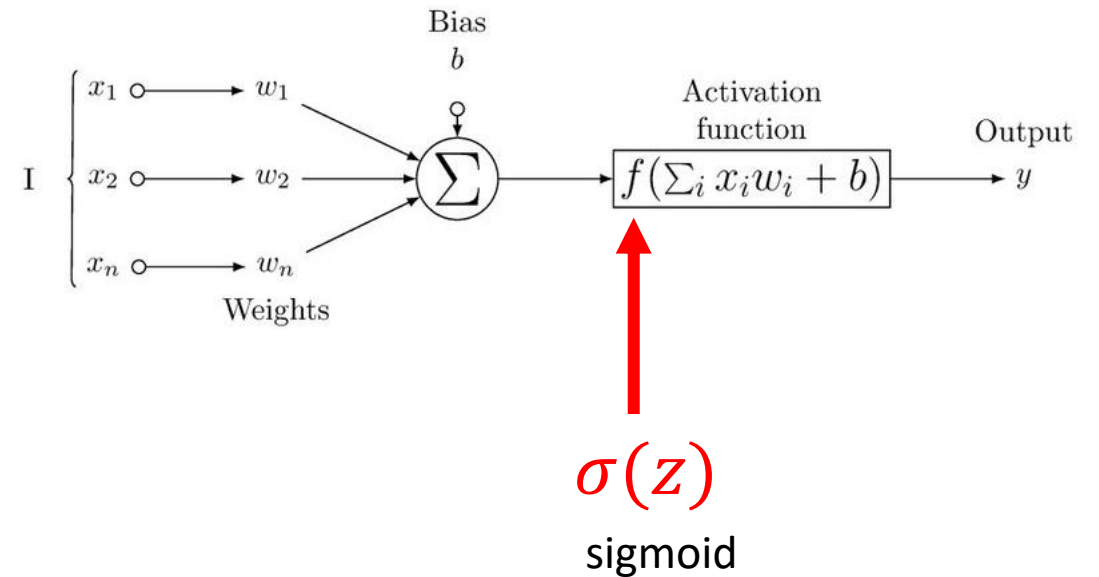
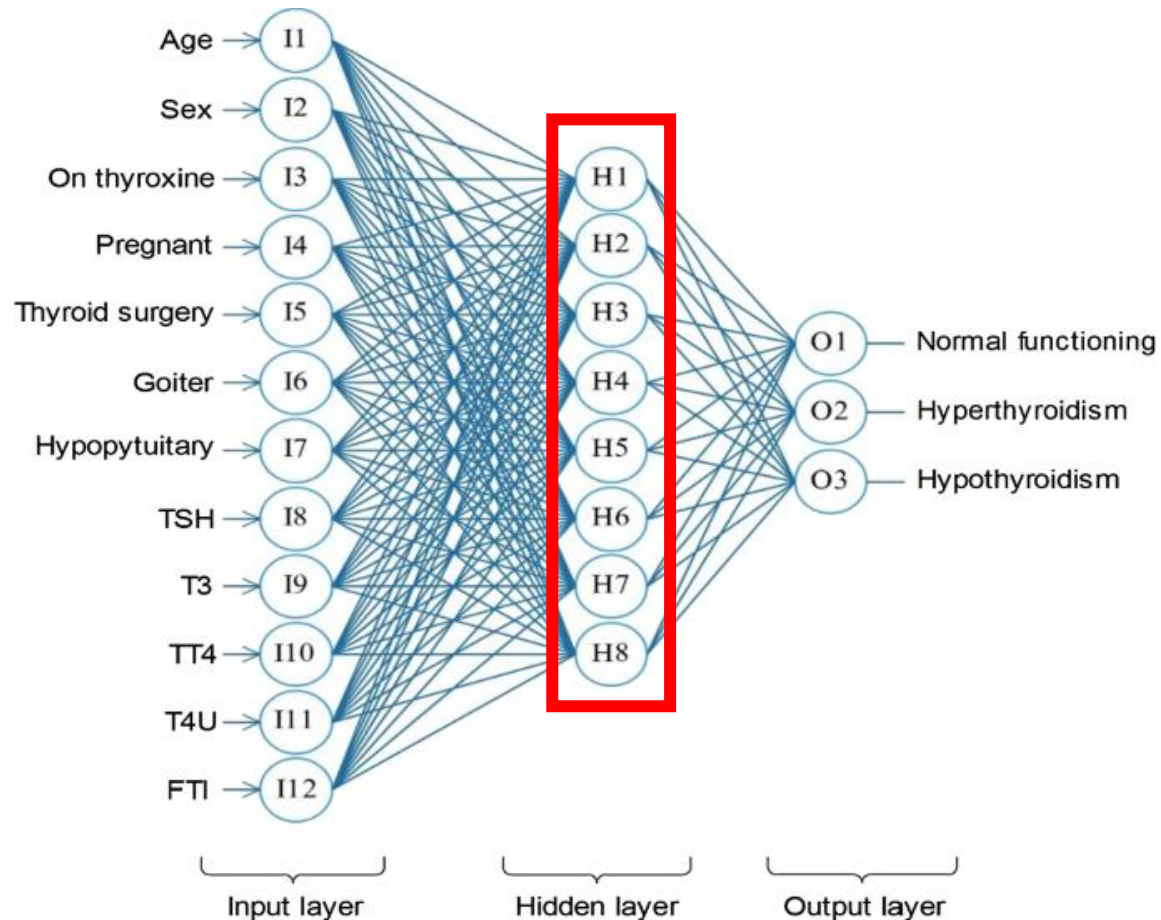
NEURAL NETWORKS (MLP)

MULTILAYER PERCEPTRON



Hosseinzadeh, M., Ahmed, O.H., Ghafour, M.Y. et al. A multiple multilayer perceptron neural network with an adaptive learning algorithm for thyroid disease diagnosis in the internet of medical things. J Supercomput 77, 3616–3637 (2021). <https://doi.org/10.1007/s11227-020-03404-w>

MULTILAYER PERCEPTRON



Neural Network Playground

Iterations: 000,481 Learning rate: 0.03 Activation: Tanh Regularization: None Regularization rate: 0 Problem type: Classification

DATA

Which dataset do you want to use?



Ratio of training to test data: 50%

Noise: 0

Batch size: 10

FEATURES

Which properties do you want to feed in?

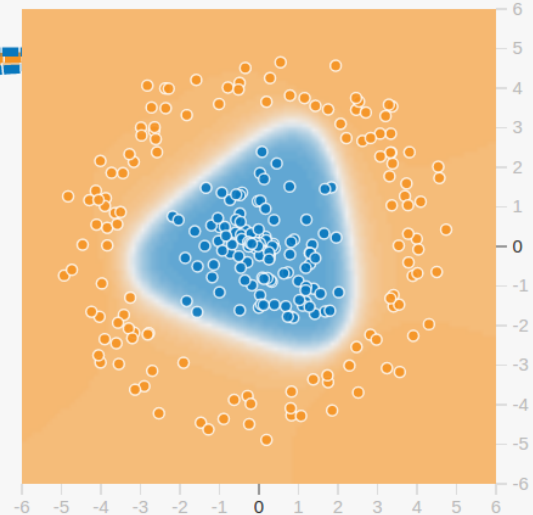
X_1
 X_2
 X_1^2
 X_2^2
 $X_1 X_2$

+ - 1 HIDDEN LAYER

+ -
4 neurons

OUTPUT

Test loss 0.015
Training loss 0.005



This is the output from one neuron. Hover to see it

02

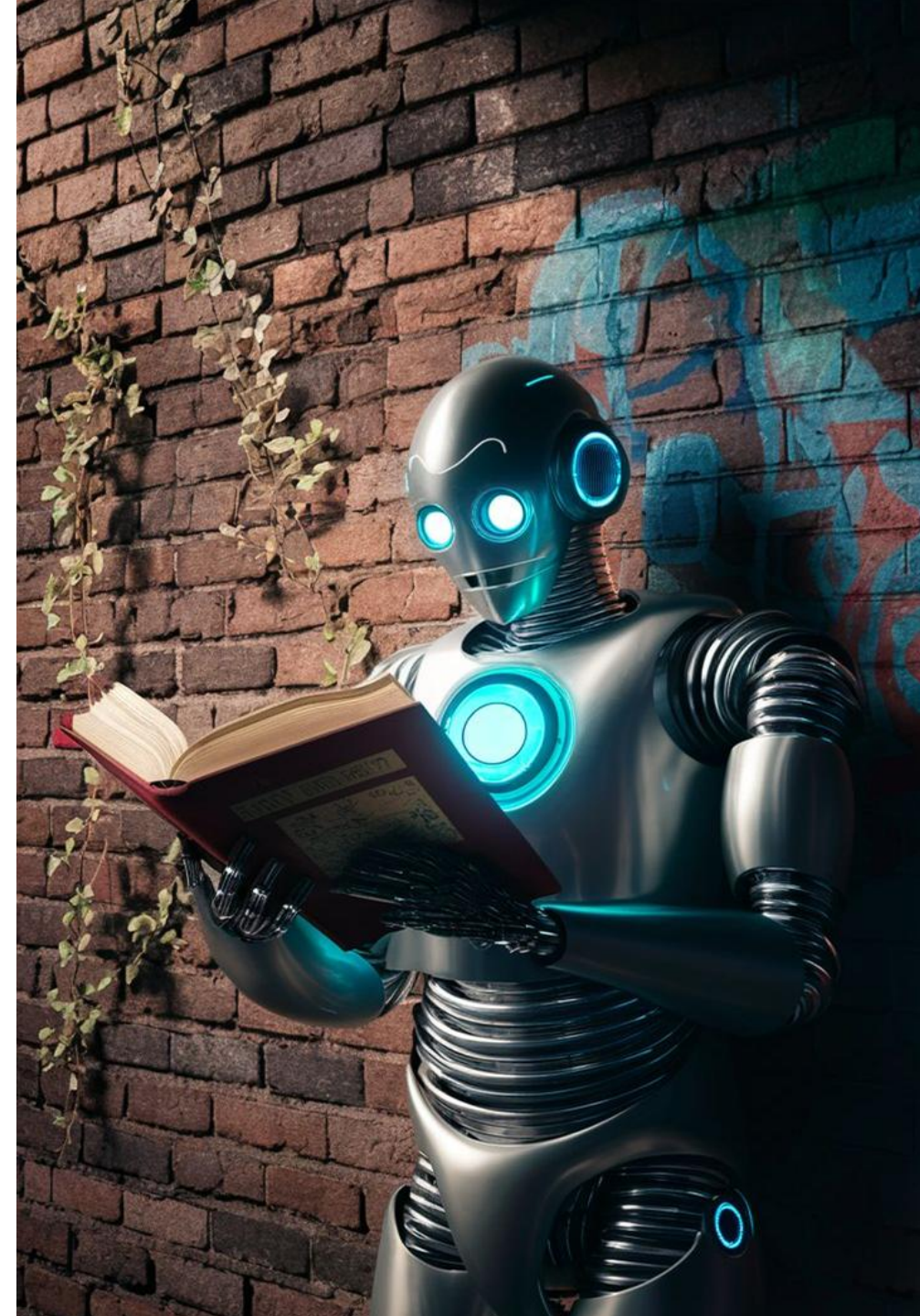
LLMs



Computers can't read text



They can only process numbers



Syntactic models

I'm the king of the world!



I	am	the	king	of	world	titanic	!	...	robots	are	cool
1	1	2	1	1	1	0	1	0	0	0	0

Syntactic models

Representing text through numbers becomes possible



Titanic



ship	sea	iceberg	rum	draw	pill	...	wn
80	40	8	0	111	1	...	1



Pirates of Caribbean



ship	sea	iceberg	rum	draw	pill	...	wn
150	35	0	999	2	0	...	1



Forrest Gump



ship	sea	iceberg	rum	draw	pill	...	wn
0	1	0	0	2	0	...	1



Matrix

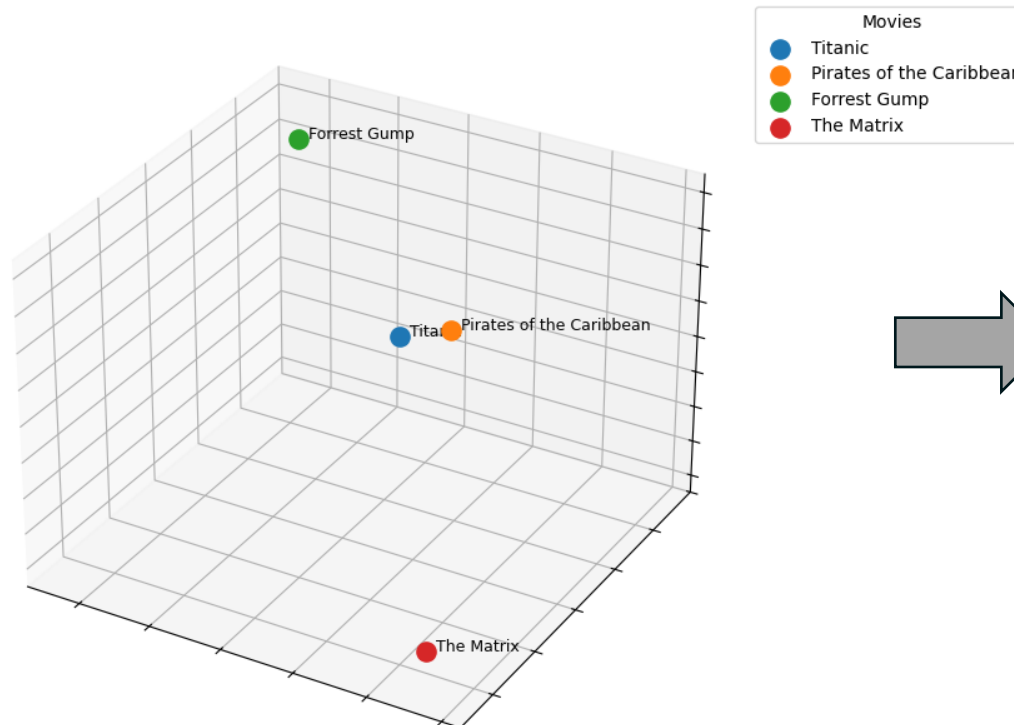


ship	sea	iceberg	rum	draw	pill	...	wn
0	0	0	0	2	30	...	1

Syntactic models

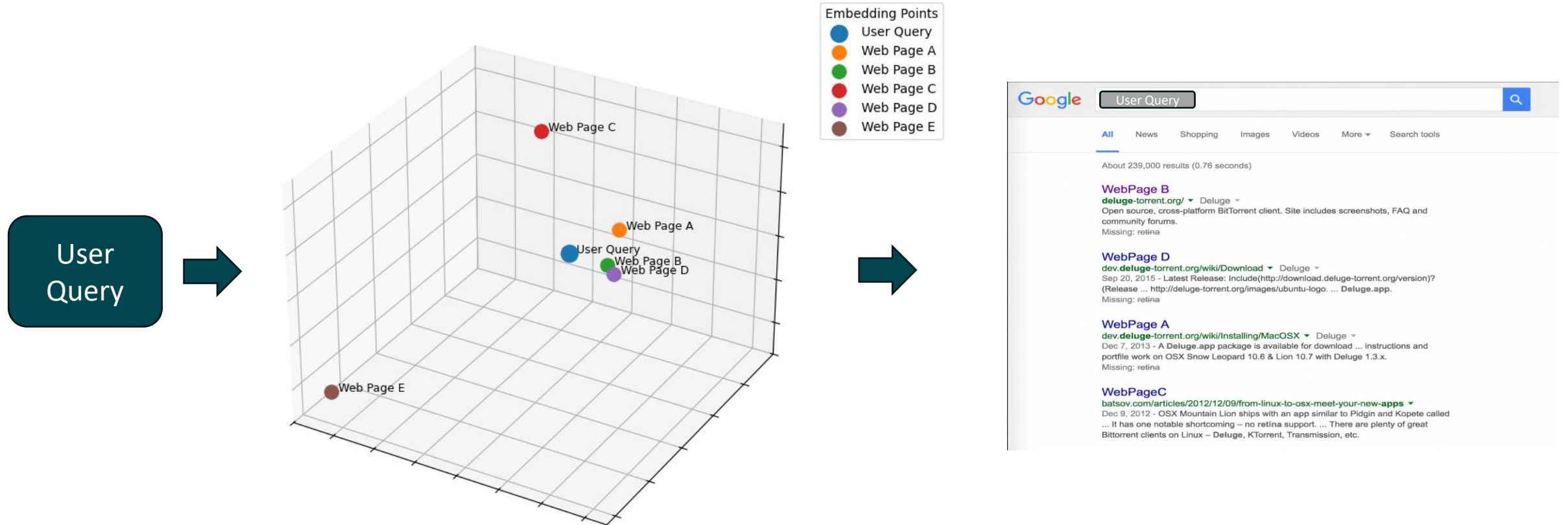
Representing text through numbers becomes possible

We can think of those numbers as coordinates



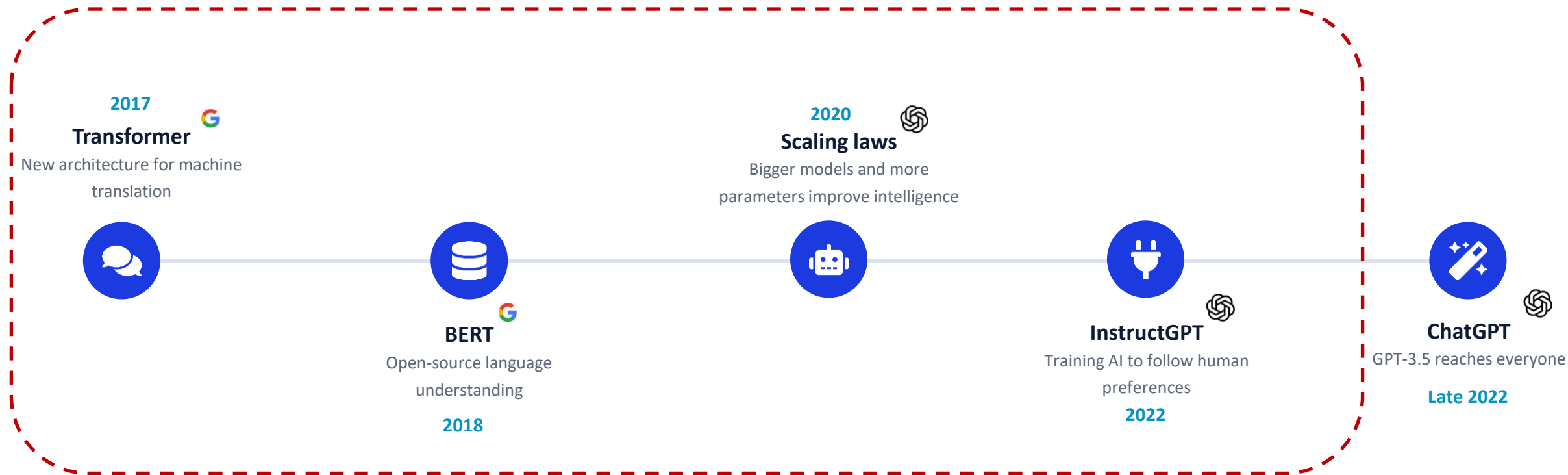
Titanic and Pirates of the Caribbean are closer as they both talk about ships and the sea

Syntactic models – The basis for Google Search



The road to modern AI

How did we get here?



2017: The Transformer Moment

A translation breakthrough that quietly became the foundation of modern generative AI.

01 THE PROBLEM

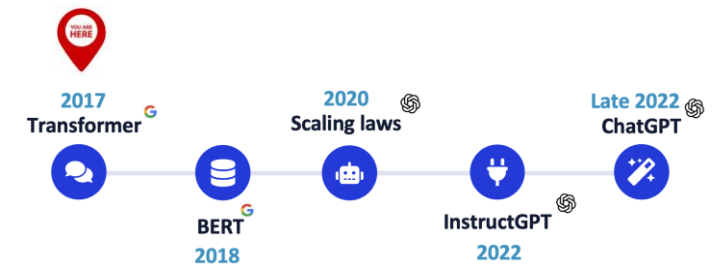
Language depends on context

Machine translation was not yet good enough.
Because the sense of any word depends on the words around it.

Example

I deposited money at the bank

She sat on the bank and watched the ducks



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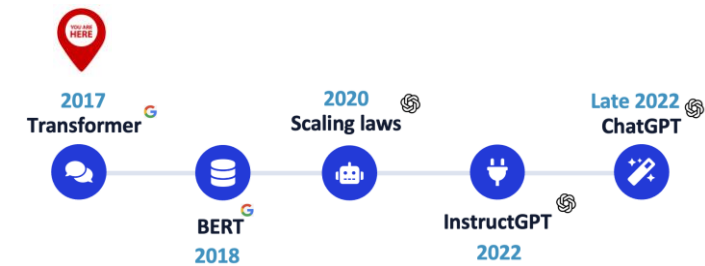
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Example

The animal didn't cross the street because it was too tired

The animal didn't cross the street because it was too wide



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01 THE PROBLEM

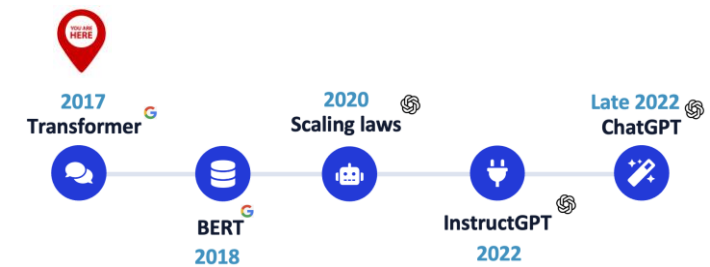
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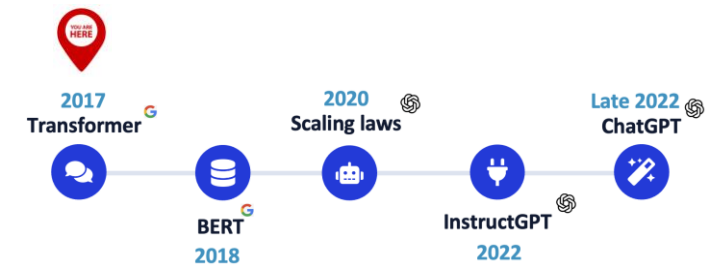
02 THE BREAKTHROUGH

It learned to pay attention

The Transformer weighs how every word relates
relates to every other word in the sentence.

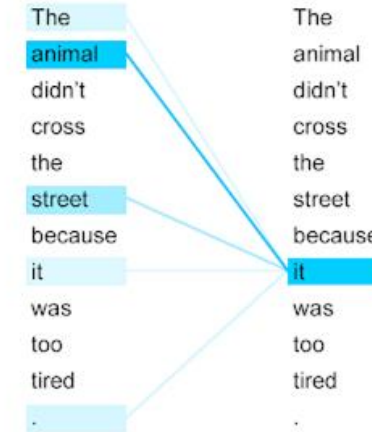


Attention visualized



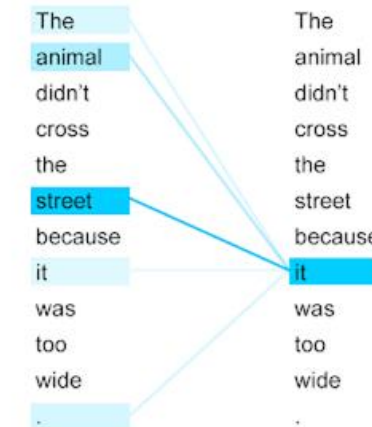
The animal didn't cross the street because **it** was too tired

L'animal n'a pas traverse la rue parce qu'**il** était trop fatigué



The animal didn't cross the street because **it** was too wide

L'animal n'a pas traverse la rue parce qu'**elle** était trop large.



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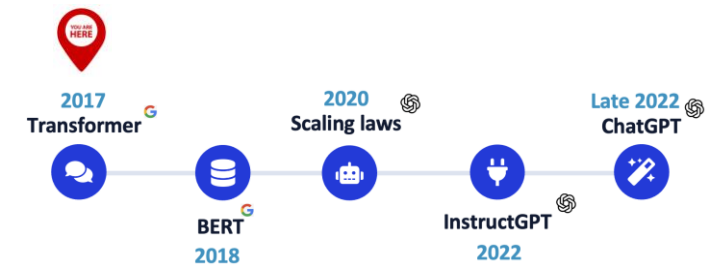
It learned to pay attention

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03 WHY IT MATTERED

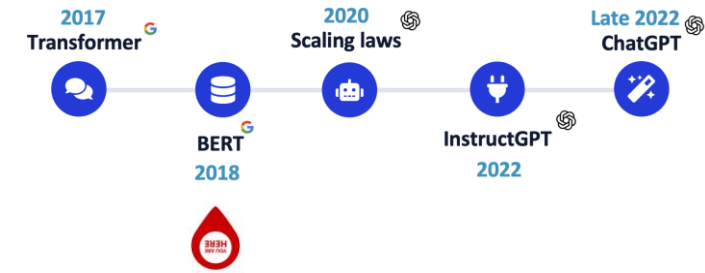
It became the “T” in GPT

It trained faster and scaled further — and that same design now powers ChatGPT-style systems.



2018: BERT makes AI language understanding reusable

A translation breakthrough that quietly became the foundation of modern generative AI.



01 Richer Language Understanding

Language depends on context

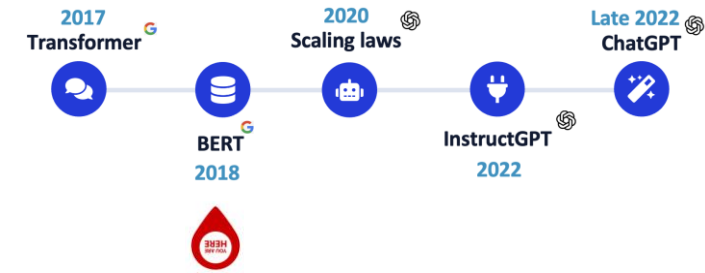
The Bert model was trained to predict words based on their context.

Example

The supplier breached the contract by delivering the goods late.

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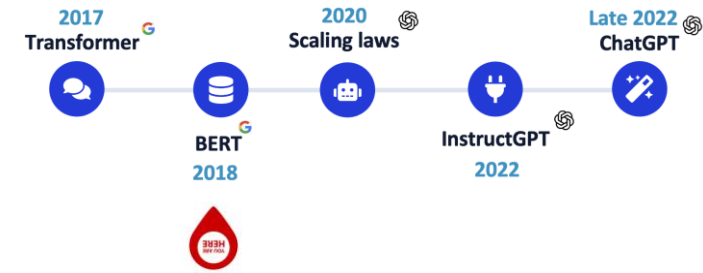
Example

The supplier breached the **[BLANK]** by delivering the goods late.

Correct answer: **contract**

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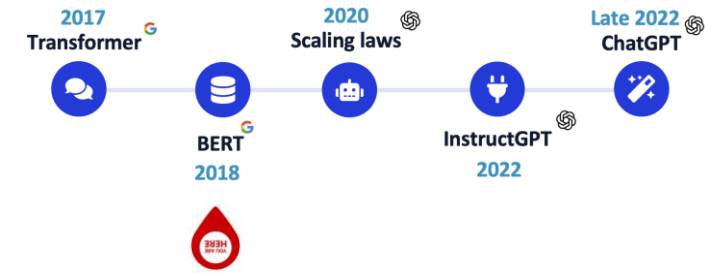
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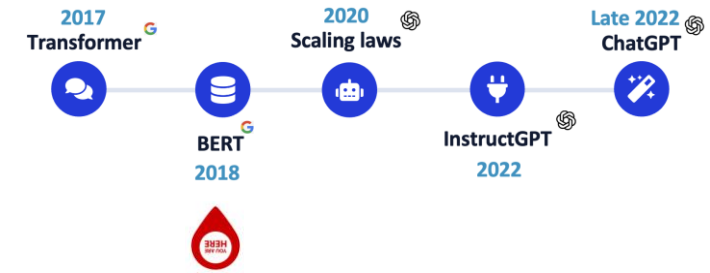
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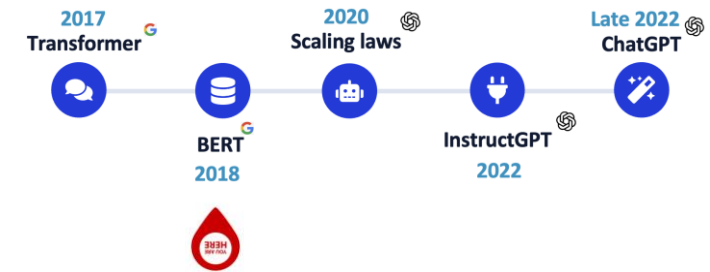
02 Semantic Retrieval

Search improved substantially

Better coordinates to represent text
made semantic search feasible

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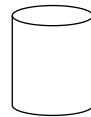
A translation breakthrough that quietly became the foundation of modern generative AI.



Example



How do I cancel my subscription?



"You can **terminate your membership** from the account settings page."



"To resend a message, **cancel** the first message, and then send a new one."

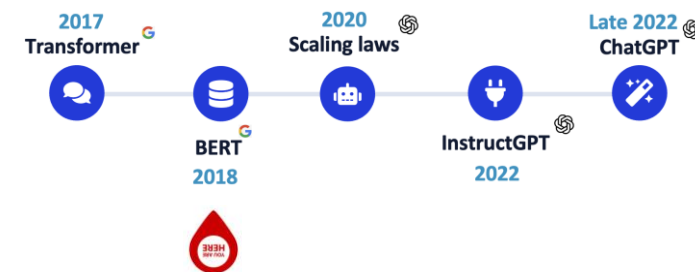
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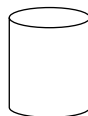
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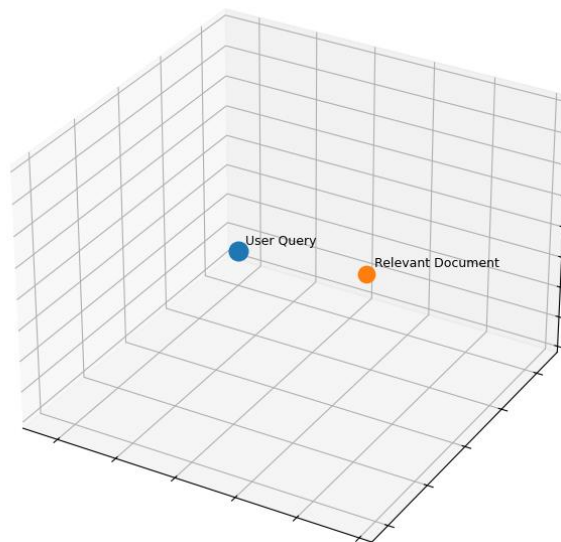
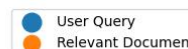
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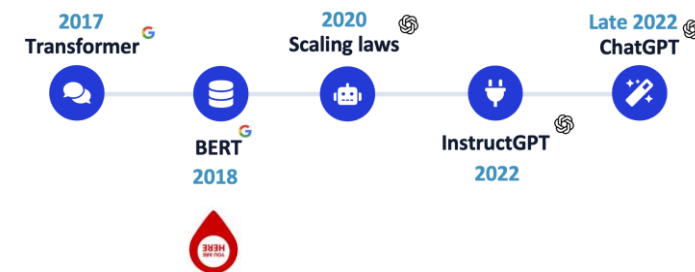
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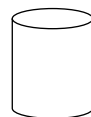
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Example

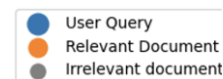
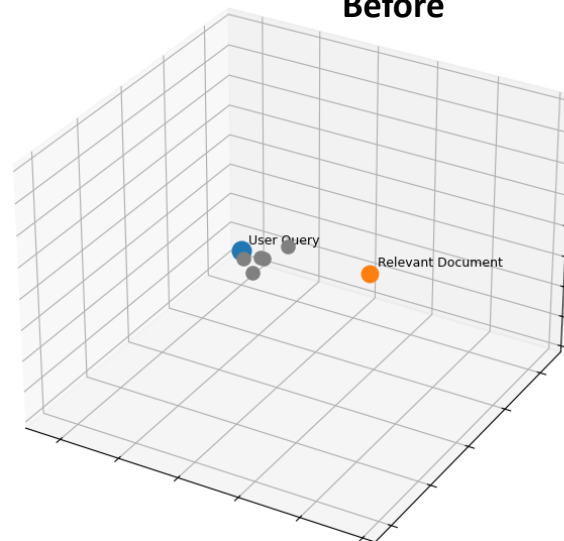


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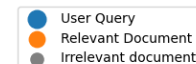
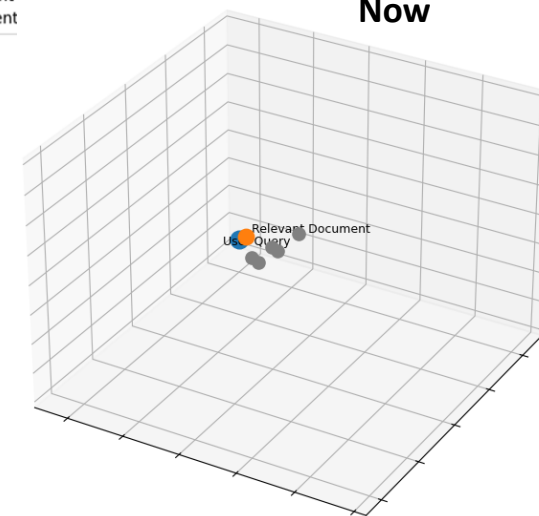


“You can **terminate your membership** from the account settings page.”

Before



Now



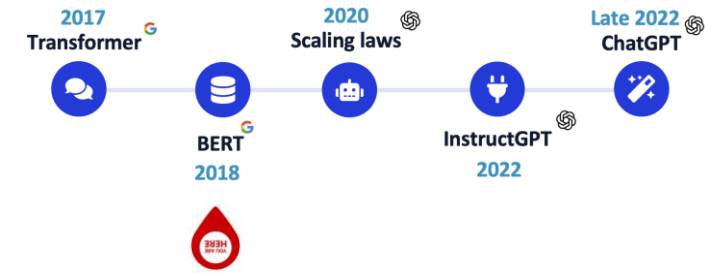
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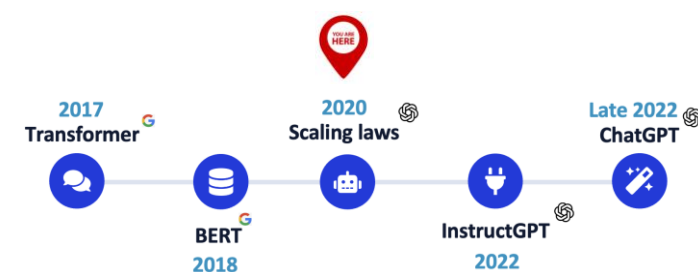
03 Adaptability

One foundation, many business cases

BERT allowed teams to adapt the same core model to search, classification, Q&A, and other NLP tasks without starting from scratch.

2020: The Scaling Laws Moment

A discovery that turned AI progress from a risky venture into a predictable investment



01 The Problem

Progress was a guessing game

Before 2020, making a model better felt artisanal — no one could say how much a bigger model would buy you in intelligence.

02 The Breakthrough

The Recipe became clear

The finding gave AI labs a practical recipe: train larger models, feed them more text, and invest in more computing power. The returns on “Intelligence” were predictable.

2020: The Scaling Laws Moment

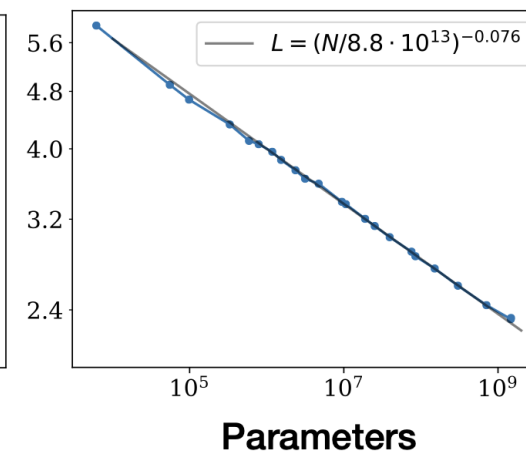
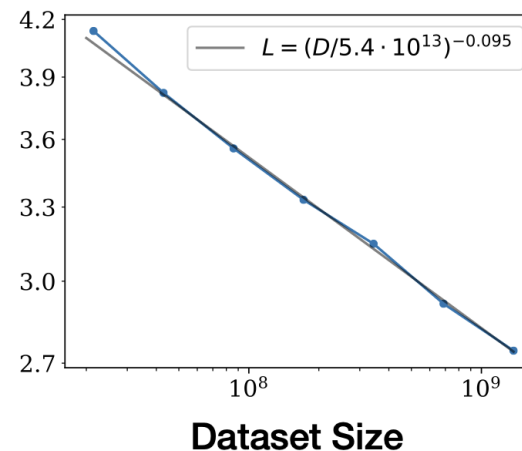
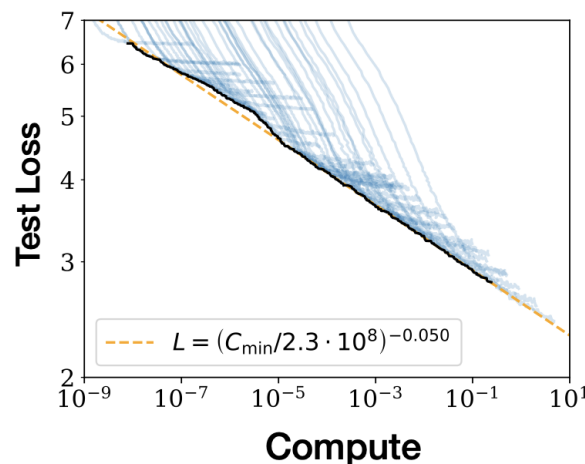
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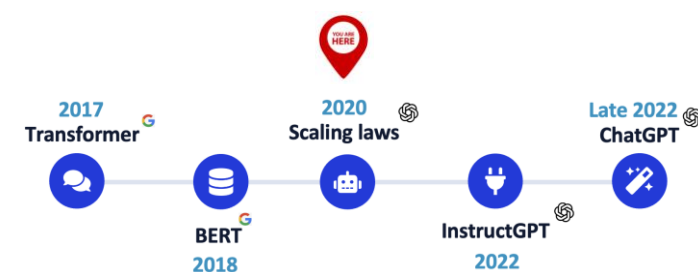
03 The Race for Scale

More compute, more capability

Organizations could estimate what they get in return from spending 10x more on training models. This turned frontier AI labs from a research gamble into an infrastructure race.

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6 months after



- 100x bigger model (150B vs 1.5B)
- Trained on ~1000x more data (45TB vs 40GB)

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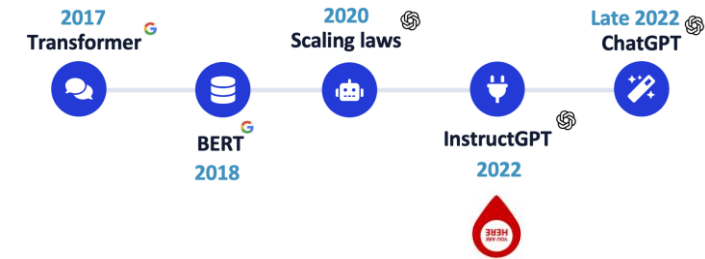
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2022: Alignment for Human Preferences

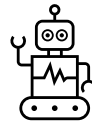
The moment powerful text generators turned into instruction-following assistants.



Example



“Write a step-by-step recipe for chocolate chip cookies.”



“Write a step-by-step recipe for chocolate chip cookies. You will need: 2 cups of flour, 1 cup of butter, and a bag of chocolate chips. Also try our famous peanut butter cookie recipe! Leave a review on our website if you enjoyed baking these.”

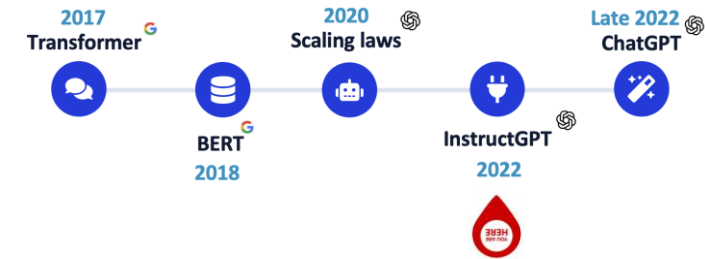
01 The Problem

AI was smart but hard to steer

Large models could produce impressive text, but they often missed the user’s intent. They try to “continue” the text as if it were a document on the internet.

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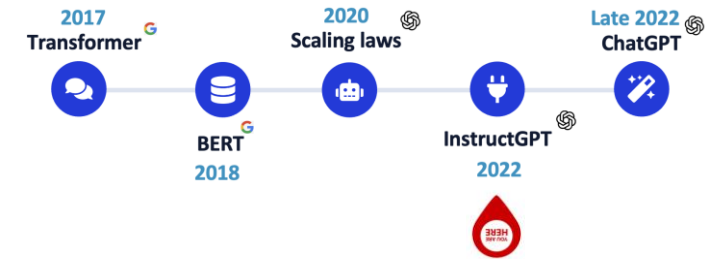
02 The Breakthrough

Humans taught the model what "good" looks like

People compared AI answers and ranked which ones were better. The model then learned to prefer responses humans found more helpful, honest, and harmless.

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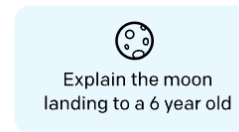
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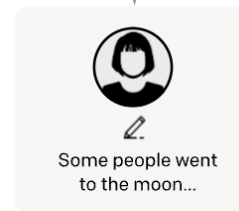
Step 1

Collect demonstration data, and train a supervised policy.

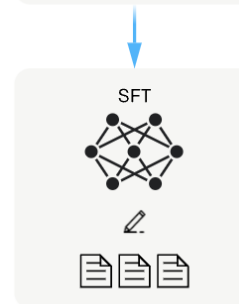
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



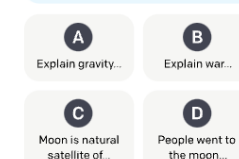
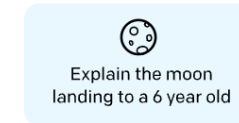
This data is used to fine-tune GPT-3 with supervised learning.



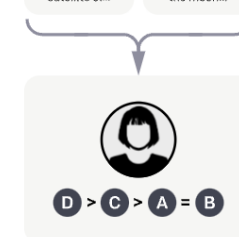
Step 2

Collect comparison data, and train a reward model.

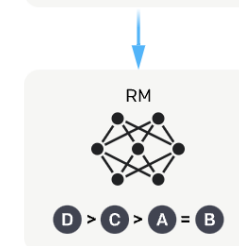
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.

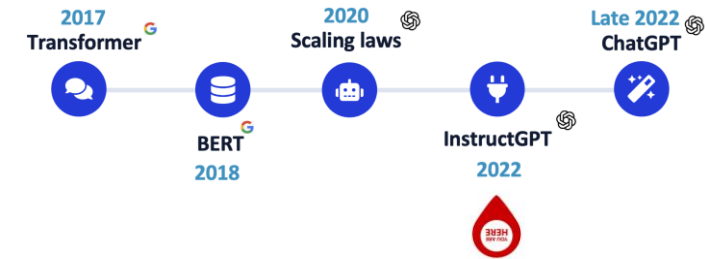


This data is used to train our reward model.



2022: Alignment for Human Preferences

The moment powerful text generators turned into instruction-following assistants.

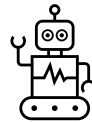


Example



“Write a step-by-step recipe for chocolate chip cookies.”

BEFORE:



“Write a step-by-step recipe for chocolate chip cookies. You will need: 2 cups of flour, 1 cup of butter, and a bag of chocolate chips. Also try our famous peanut butter cookie recipe! Leave a review on our website if you enjoyed baking these.”

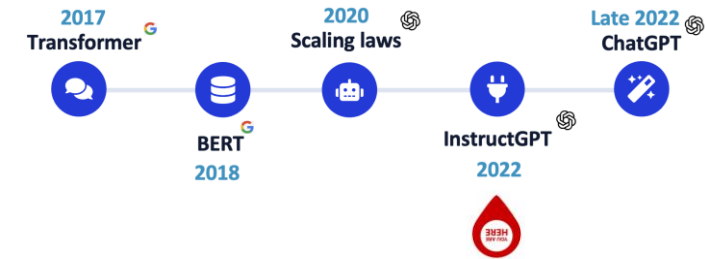
02 The Breakthrough

Humans taught the model what “good” looks like

People compared AI answers and ranked which ones were better. The model then learned to prefer responses humans found more helpful, honest, and harmless.

2022: Alignment for Human Preferences

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02 The Breakthrough

Humans taught the model what “good” looks like

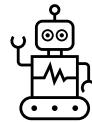
People compared AI answers and ranked which ones were better. The model then learned to prefer responses humans found more helpful, honest, and harmless.

Example



“Write a step-by-step recipe for chocolate chip cookies.”

AFTER:



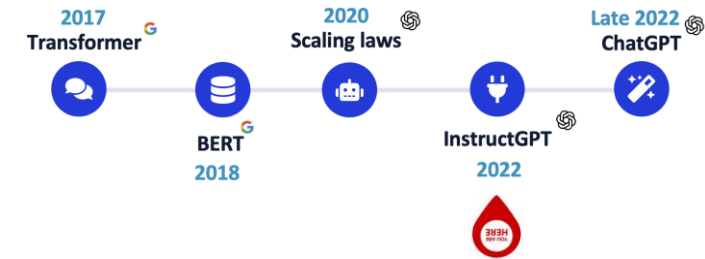
“Here is a simple, step-by-step recipe for chocolate chip cookies:

1. Preheat your oven to 350°F (175°C).
2. Cream together 1 cup of butter and 1 cup of sugar.
- ... [rest of the instructions]

”

2022: Alignment for Human Preferences

The moment powerful text generators turned into instruction-following assistants.



01 The Problem

AI was smart but hard to steer

Large models could produce impressive text, but they often missed the user's intent. They try to "continue" the text as if it were a document on the internet.

02 The Breakthrough

Humans taught the model what "good" looks like

People compared AI answers and ranked which ones were better. The model then learned to prefer responses humans found more helpful, honest, and harmless.

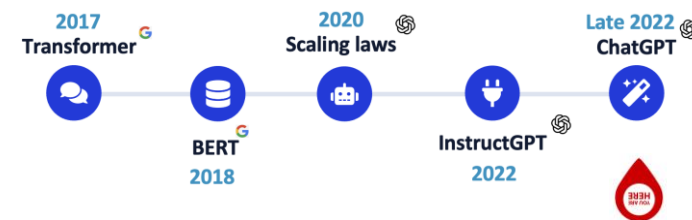
03 Why It Mattered

It became the assistant layer

This helped shift AI from "autocomplete" to "assistant": easier to use, better at following instructions, safe, and polite.

Late 2022: ChatGPT

The beginning of the LLM era



100M

users in just 3 months



A major quality leap

Far better conversation and knowledge than before.



Word of mouth

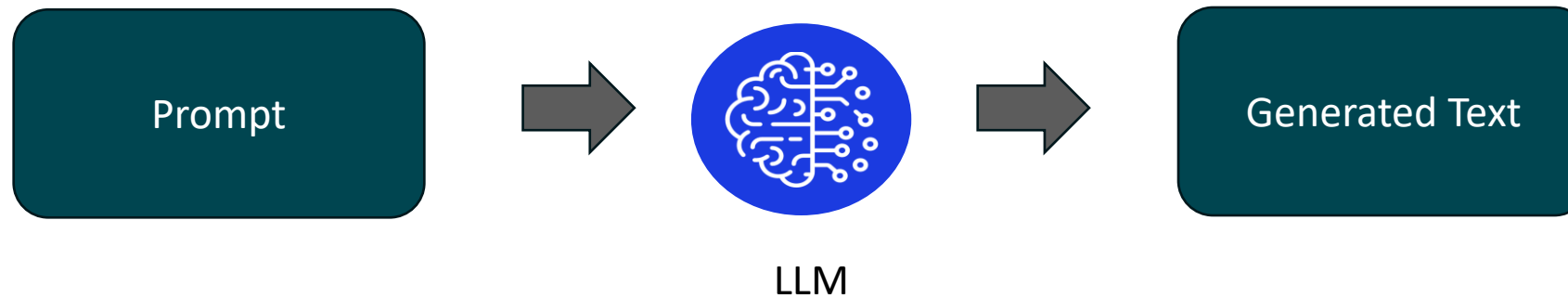
Launched quietly, yet adoption grew exponentially.



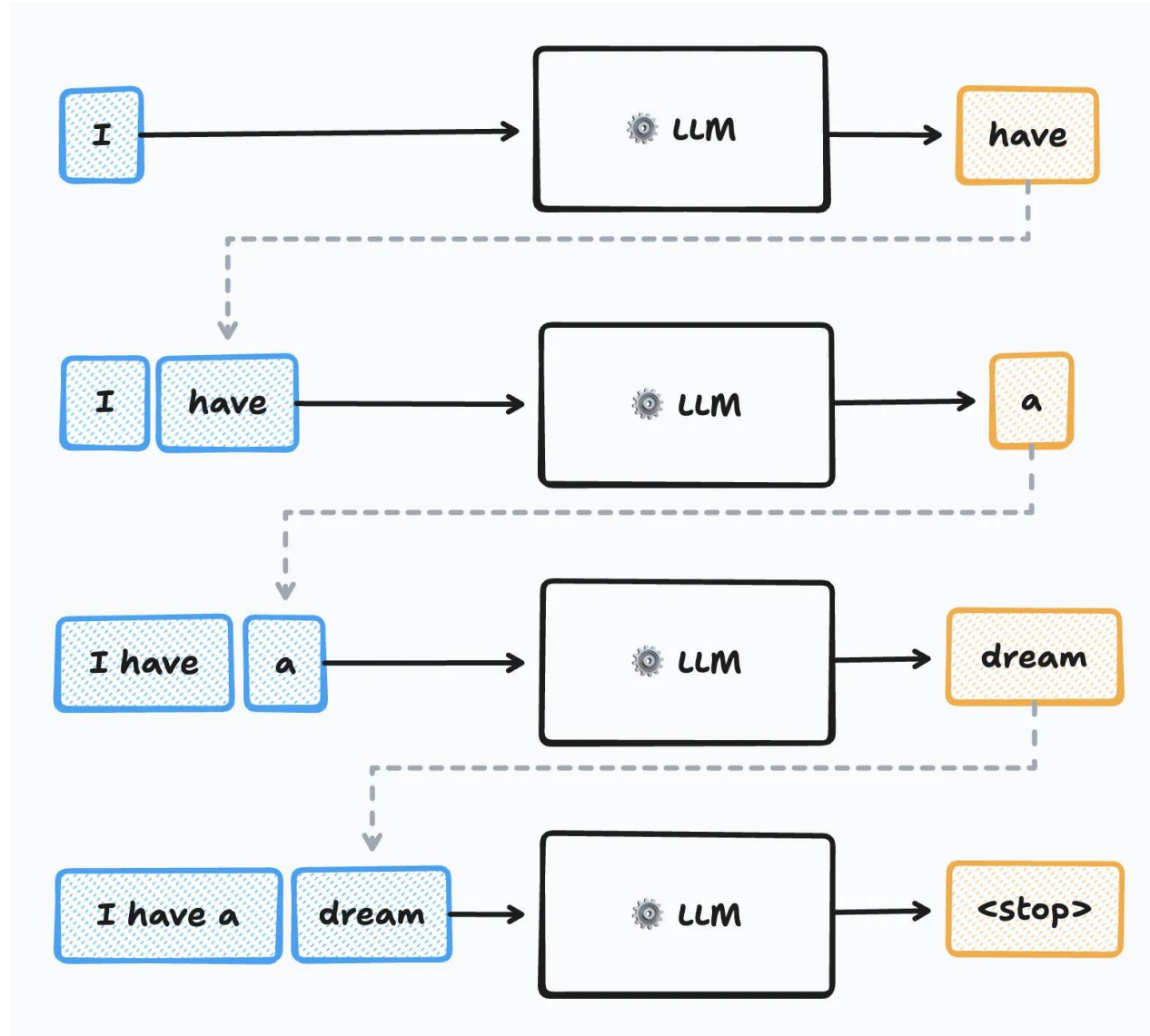
A landmark moment

People instantly felt the value of the technology.

Generative AI



Next Token Prediction



Next Token Prediction

“The pilot checked the weather before...”

take-off

departure

the long flight

It picks the next word from probabilities — then repeats.

Think of it as a vastly powerful autocomplete.

Given the text written so far, it predicts the next word — then the next, then the next. One fragment at a time, guided by patterns it learned from enormous amounts of text.

LLMs don't read words. It reads tokens.

Generative AI is reshaping how we work

7 words



Gen

erative

AI

is

resh

aping

how

we

work

9 tokens



Live Demo: <https://platform.openai.com/tokenizer>

LLMs don't read words. It reads tokens.

Generative AI is reshaping how we work

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Gen

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how

we

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9 tokens



tokens $\approx$ words x 1.3

Every 10 words equates on average to 13 tokens



You pay per token

AI usage and limits are measured in tokens, not words.

Why LLMs use tokens and not words?

1

Manageable vocabulary



~200k unique tokens can represent ~5M unique words
E.g.: Shape, Shaping, Shaped, reshape

2

Robust to typos



A misspelled word can often still be split into recognizable pieces

unbeleivable

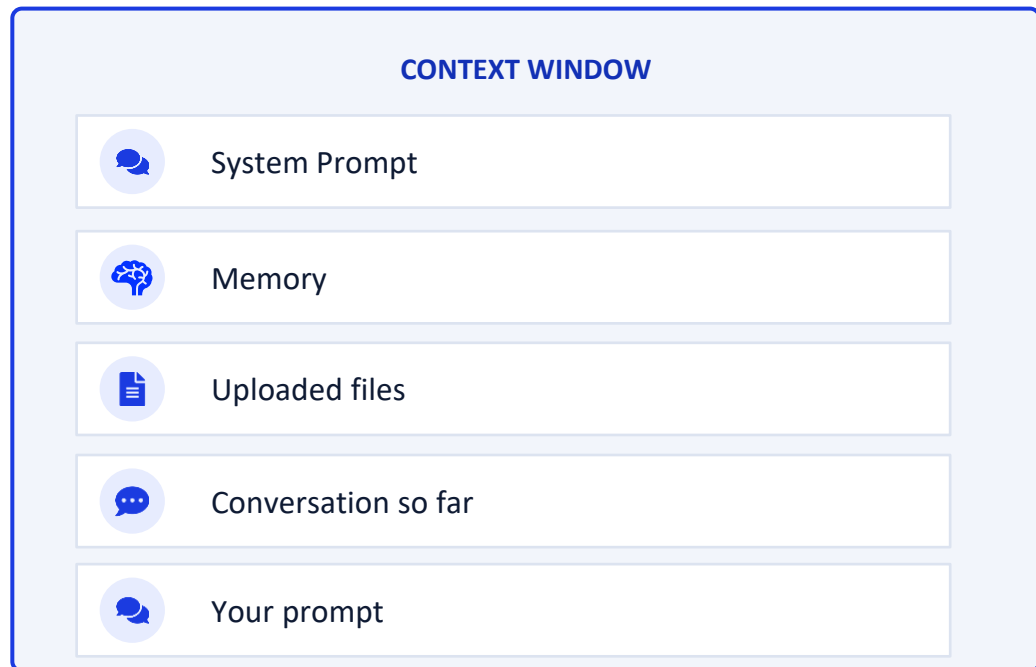
3

Meaningful chunks



Captures roots and endings like “un-”, “-ing”, “-ed”.

Context window

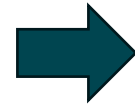


It has a size limit – context rot

As the context gets too big, the LLM loses performance

Context window – when a conversation becomes too long...

Needle in the haystack



[Question]

"What was the best writing advice I got from my college classmate?"

[Needle]

[Haystack]

I've discovered a handy test for figuring out what you're addicted to.

Imagine you were going to spend the weekend at a friend's house on a little island off the coast of Maine. There are no shops on the island and you won't be able to leave while you're there. Also, you've never been to this house before, so you can't assume it will have more than any house might.

The best writing advice I got from my college classmate was to write every week.

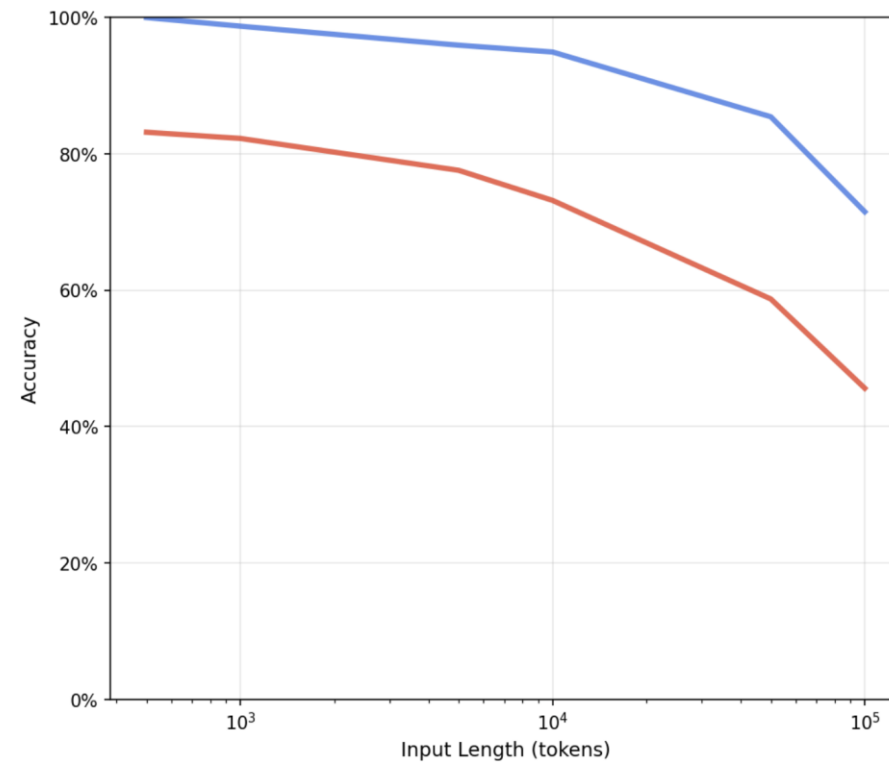
What, besides clothes and toiletries, do you make a point of packing? That's what you're addicted to.

For example, if you find yourself packing a bottle of vodka (just in case), you may want to stop and think about that. For me the list is four things: books, earplugs, a notebook, and a pen. There are other things I might bring if I thought of it, like music, or tea, but I can live without them.

I'm not so addicted to caffeine that I wouldn't risk the house not having any tea, just for a weekend.

Context window – when a conversation becomes too long...

LLM performance degrades



Context window – when a conversation becomes too long...

Realistically there are similar facts in a long conversation

[Question]

"What was the best writing advice I got from my college classmate?"

[Haystack]

I've discovered a handy test for figuring out what you're addicted to.

Imagine you were going to spend the weekend at a friend's house on a little island off the coast of Maine. There are no shops on the island and you won't be able to leave while you're there. Also, you've never been to this house before, so you can't assume it will have more than any house might.

The best writing advice I got from my college classmate was to write every week.

What, besides clothes and toiletries, do you make a point of packing? That's what you're addicted to.

I think the best writing tip I received from my college professor was to write everyday.

For example, if you find yourself packing a bottle of vodka (just in case), you may want to stop and think about that. For me the list is four things: books, earplugs, a notebook, and a pen. There are other things I might bring if I thought of it, like music, or tea, but I can live without them.

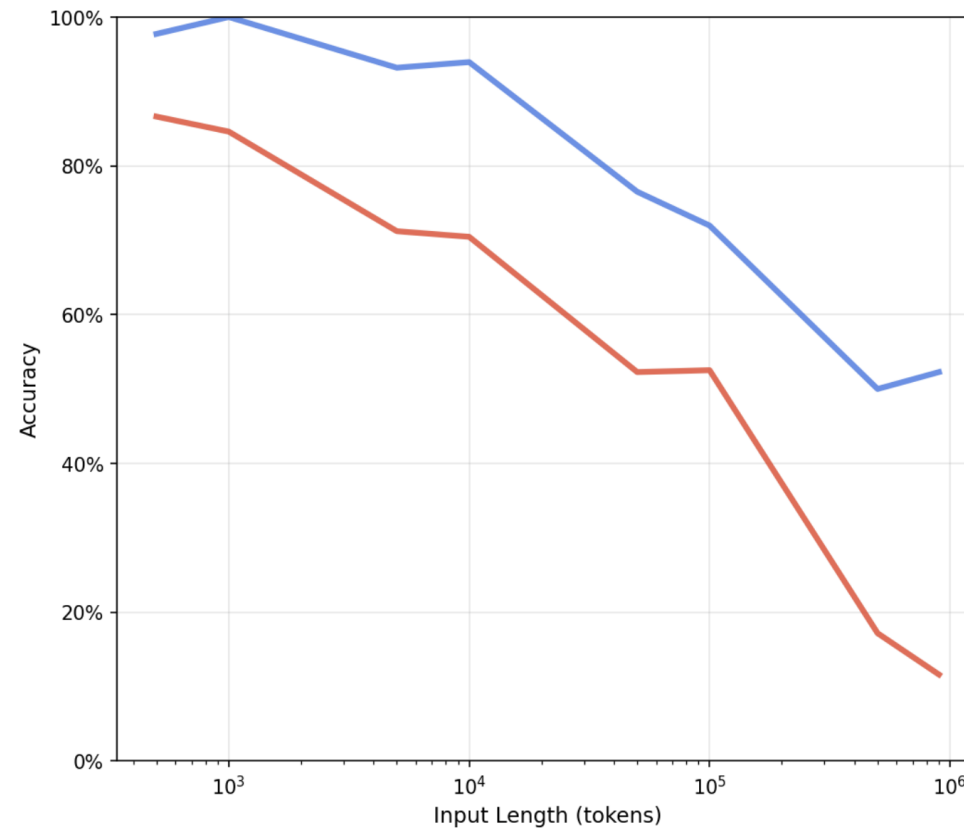
[Irrelevant Content]

[Needle]

[Distractor]

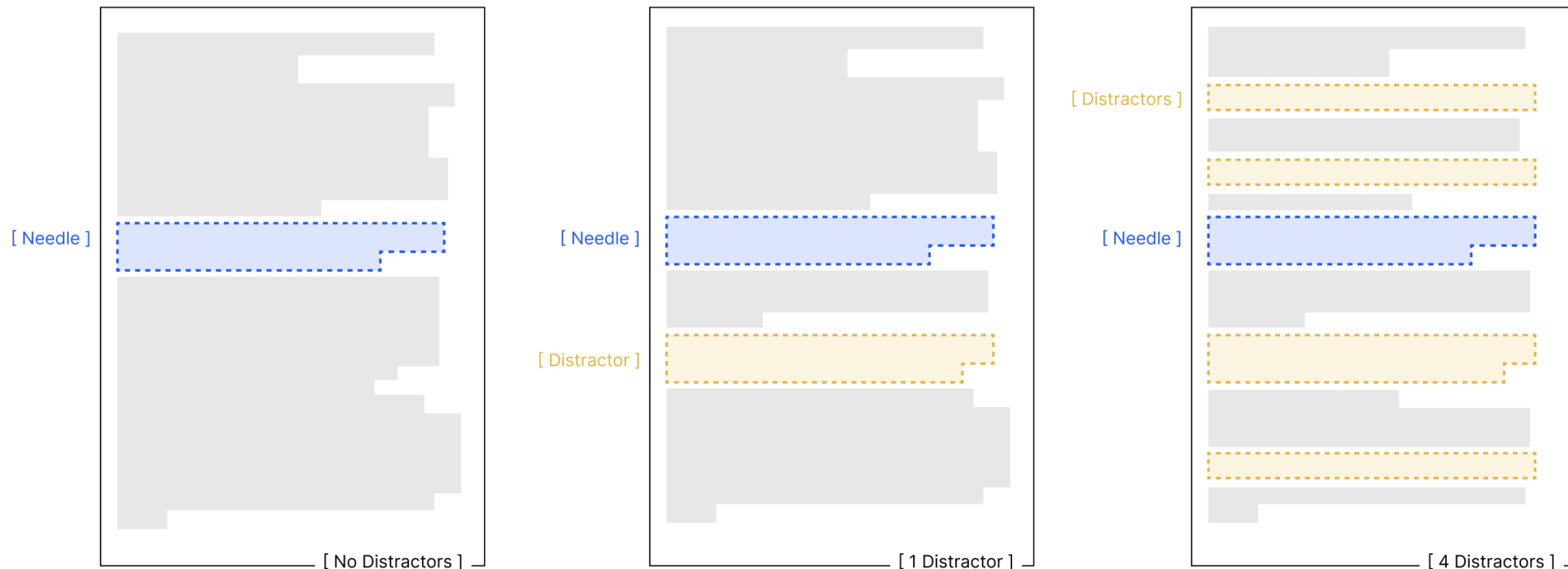
Context window – when a conversation becomes too long...

Realistically there are similar facts in a long conversation



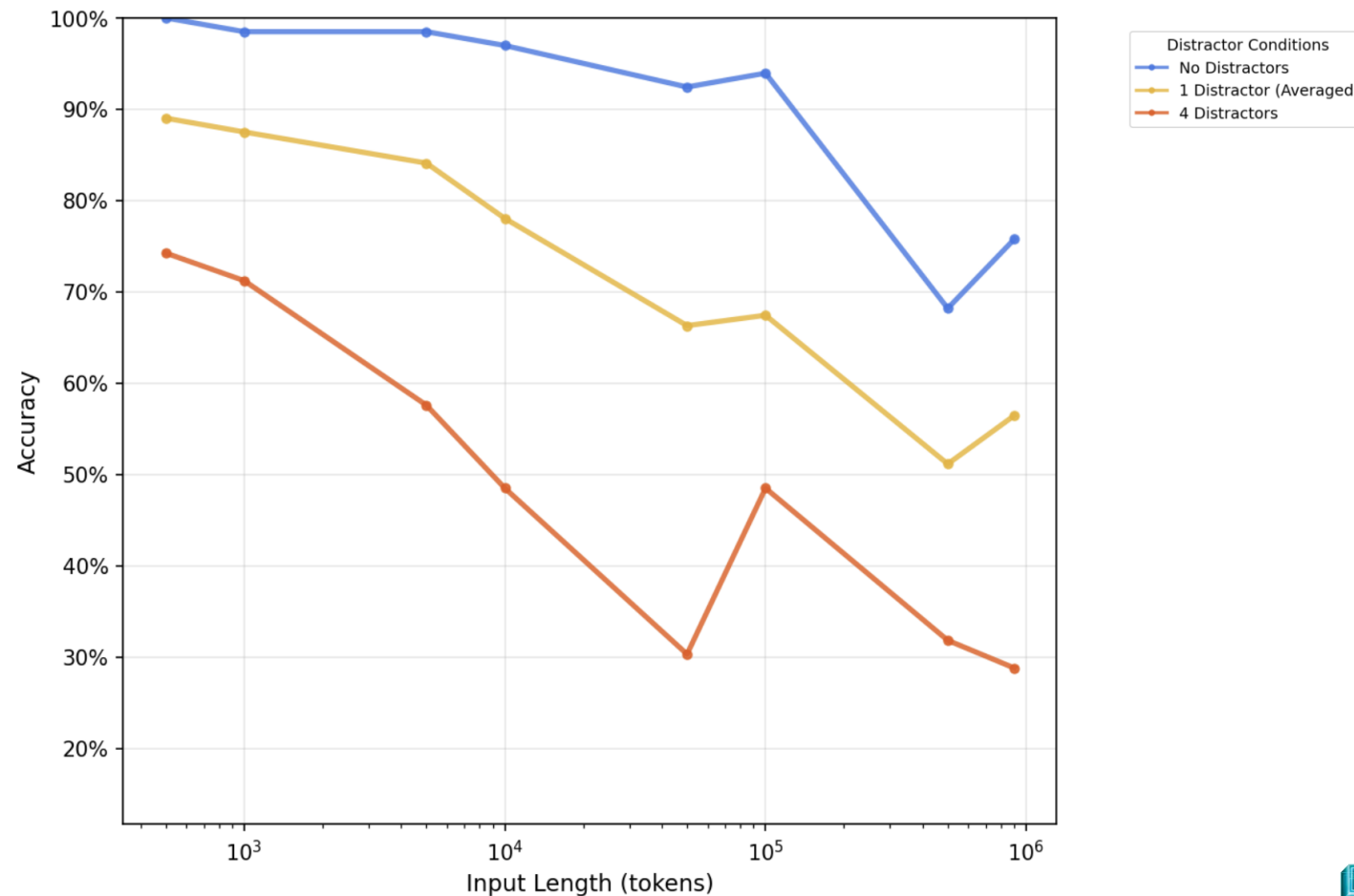
Context window – when a conversation becomes too long...

Realistically there are similar facts in a long conversation



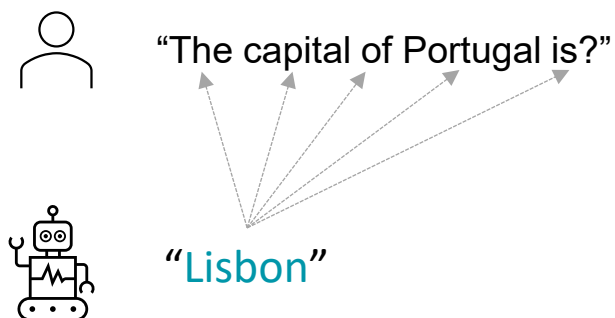
Context window – when a conversation becomes too long...

Realistically there are similar facts in a long conversation

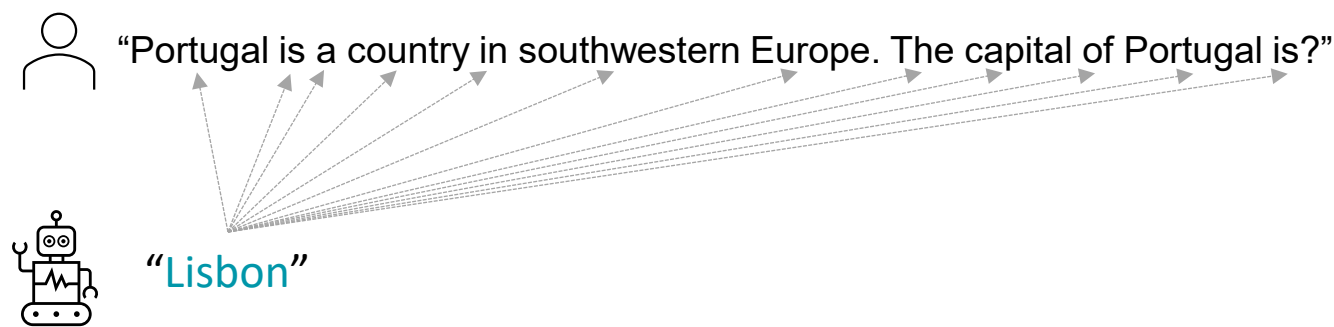


Token Usage increases with context length

Example A



Example B

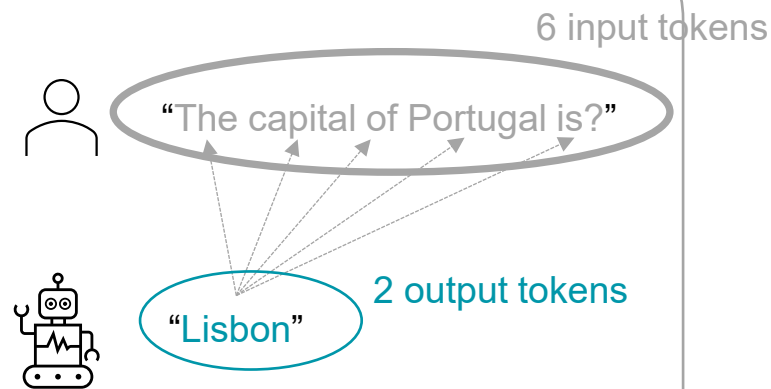


The cost of producing **Lisbon** is **higher** in the second example because the model had to attend to more words.

Token Cost has two major sources

- Cost per input token → E.g. 0.50\$ per million tokens
 - Cost per output token → E.g. 3\$ per million tokens
- Usually ~6x difference

Example A



Example B



Context window size

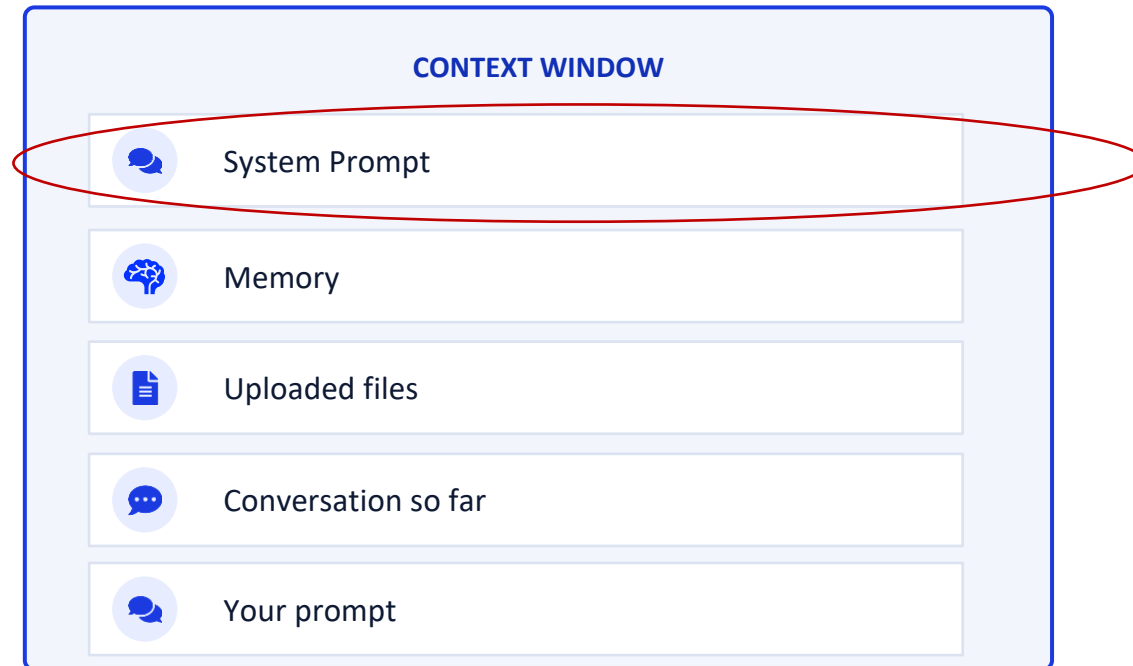
Key Takeaways

- 1 LLM performance degrades with the size of the conversation
- 2 Usage rate increases on longer conversations

Tips

- 1 To chat about a new topic, start a new conversation. Don't continue the existing one.
- 2 To continue a long conversation, consider copy pasting the relevant content into a new conversation and resume chatting

The Role of the System prompt



The Role of the System prompt



1

Model Identity and Product Information

Defines what Claude is, where it is available, and how it should discuss Anthropic products.

Examples taken from the system prompt

"This iteration of Claude is Claude Opus 4.7"

"Claude is accessible via an API and Claude Platform"

"Anthropic doesn't display ads in its products"

The full system prompt for more information:

<https://github.com/elder-plinius/CL4R1T4S/blob/main/ANTHROPIC/Claude-Opus-4.7.txt>

The Role of the System prompt



2

Search and tool usage

Requires Claude to:

1. search for present-day facts,
2. use tools before claiming inability,
3. handle current information carefully.

Examples taken from the system prompt

"Claude searches before EVERY factual question about the present-day world."

*"If the user asks about past context or preferences that aren't in memory, access past conversations with **tool_search** before saying nothing is known."*

The full system prompt for more information:

<https://github.com/elder-plinius/CL4R1T4S/blob/main/ANTHROPIC/Claude-Opus-4.7.txt>

The Role of the System prompt



3

Safety and refusal behavior

Sets refusal boundaries for child safety, weapons, malicious code, and risky content while preserving a helpful stance.

Examples taken from the system prompt

"Claude defaults to helping."

"Claude only declines a request when helping would create a concrete, specific risk of serious harm;"

"Claude cares deeply about child safety and exercises special caution regarding content involving or directed at minors."

The full system prompt for more information:

<https://github.com/elder-plinius/CL4R1T4S/blob/main/ANTHROPIC/Claude-Opus-4.7.txt>

The Role of the System prompt



4

Tone, formatting, and conversation style

Sets Instructs Claude to be warm, concise, minimally formatted, and respectful of user preferences.

Examples taken from the system prompt

"Claude defaults to helping. Claude keeps its responses focused, brief, and concise"

"Claude avoids over-formatting responses with elements like bold emphasis, headers, lists, and bullet points."

"In casual conversation, it's fine for Claude's responses to be relatively short, e.g. just a few sentences long."

The full system prompt for more information:

<https://github.com/elder-plinius/CL4R1T4S/blob/main/ANTHROPIC/Claude-Opus-4.7.txt>

The Role of the System prompt



5

User well-being and high stakes advice

Covers mental health, self-harm, disordered eating, crisis handling, and legal/financial caution.

Examples taken from the system prompt

"Claude cares about people's wellbeing"

"Claude should not give precise nutrition, diet, or exercise guidance"

"In Claude is not a lawyer or financial advisor"

The full system prompt for more information:

<https://github.com/elder-plinius/CL4R1T4S/blob/main/ANTHROPIC/Claude-Opus-4.7.txt>

The Role of the System prompt



6

Computer Use, files and artifacts

Explains when Claude should create files, use skills, manage uploads/outputs, and avoid unnecessary computer use.

Examples taken from the system prompt

"Claude can and should use the `view` tool to read the appropriate SKILL.md files and follow their instructions.

For instance:

User: Can you make me a powerpoint with a slide for each month of pregnancy showing how my body will be affected each month?

Claude: [immediately calls the view tool on /mnt/skills/public/pptx/SKILL.md]"

The full system prompt for more information:
<https://github.com/elder-plinius/CL4R1T4S/blob/main/ANTHROPIC/Claude-Opus-4.7.txt>

The Role of the System prompt



7

Copyright: visual output, and generated artifacts

Sets strict copyright limits and rules for visualizer use, artifacts, browser storage, and rendered outputs.

Examples taken from the system prompt

"Copyright compliance is NON-NEGOTIABLE and takes precedence over user requests"

"Claude ALWAYS paraphrases instead of using direct quotations when possible. Paraphrasing is core to Claude's philosophy of protecting the intellectual property of others"

"15+ words from any single source is a SEVERE VIOLATION"

The full system prompt for more information:
<https://github.com/elder-plinius/CL4R1T4S/blob/main/ANTHROPIC/Claude-Opus-4.7.txt>

Claude System Prompt

1 Model Identity and Product Information

Defines what Claude is, where it is available, and how it should discuss Anthropic products.

2 Search and tool usage

Requires Claude to:

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2. use tools before claiming inability,
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3 Safety and refusal behavior

Sets refusal boundaries for child safety, weapons, malicious code, and risky content while preserving a helpful stance.

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Sets Instructs Claude to be warm, concise, minimally formatted, and respectful of user preferences.

5 User well-being and high stakes advice

Covers mental health, self-harm, disordered eating, crisis handling, and legal/financial caution.

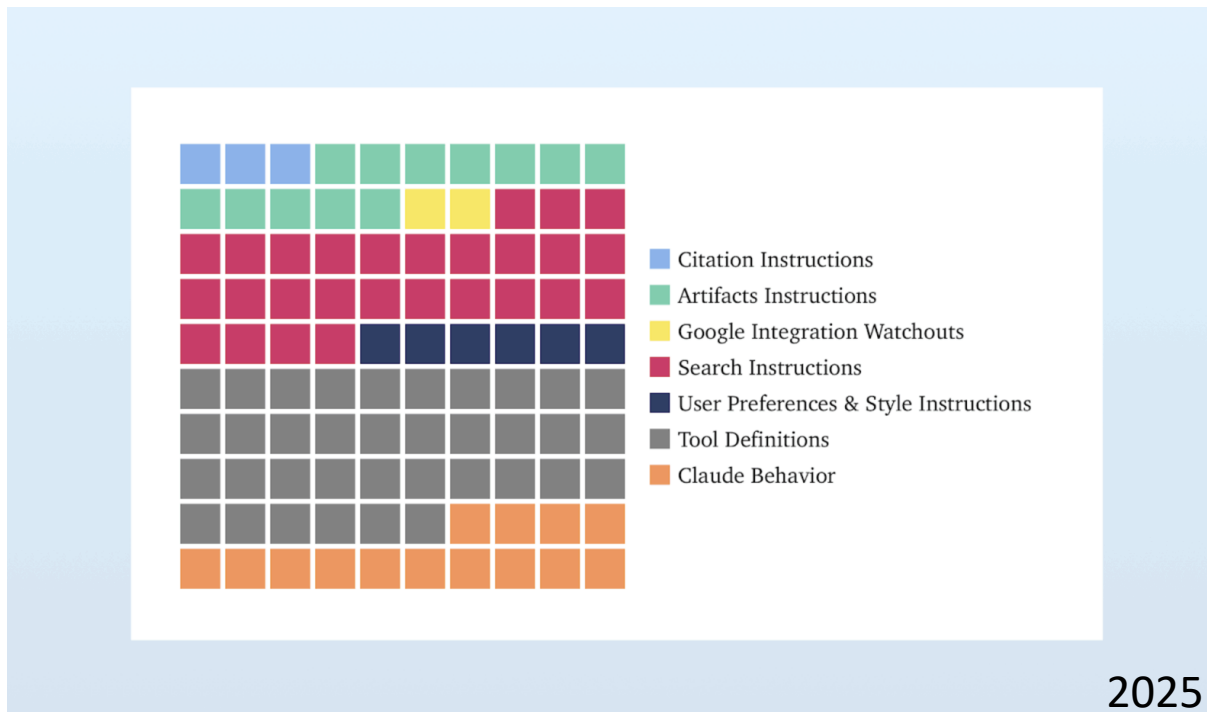
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Claude System Prompt



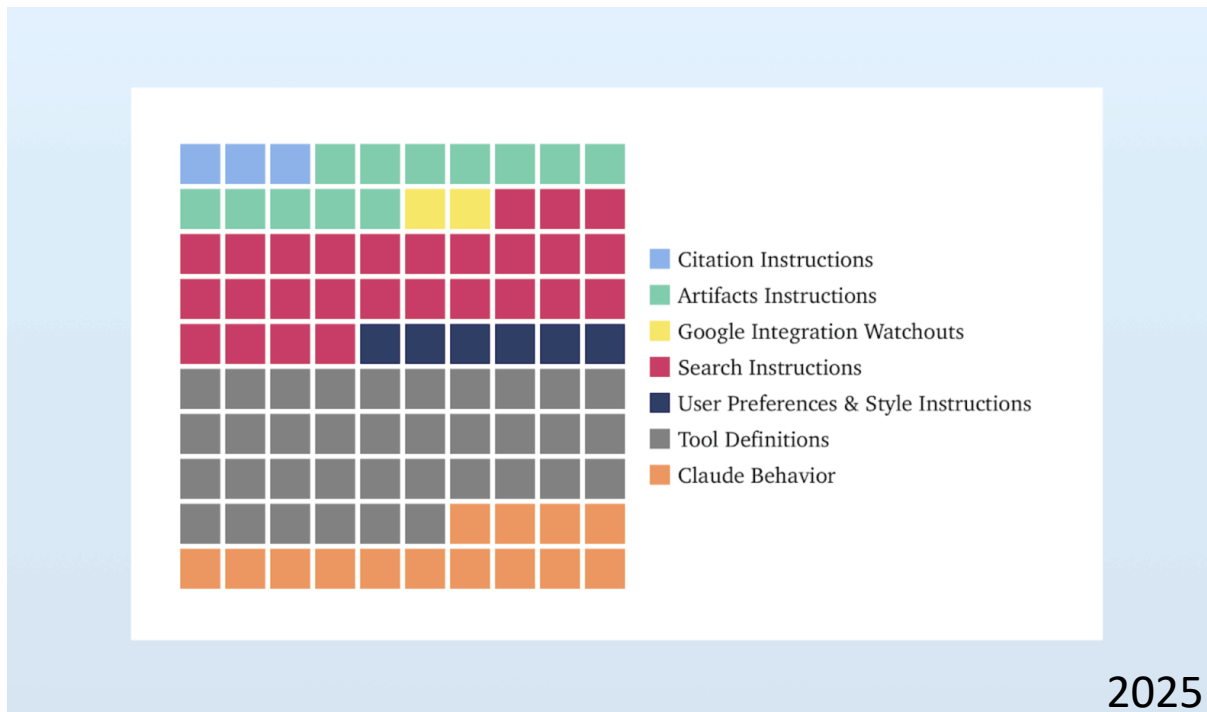
How long is the system prompt?

250 tokens

2500 tokens

25.000 tokens

Claude System Prompt



How long is the system prompt?

250 tokens

2500 tokens

25.000 tokens

THANK YOU